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**“Does Worker Wellbeing Pay Off? Employee Sentiment, Operating Performance, and
Risk-Adjusted Stock Returns in the S&P 500, 2013–2023”**

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List of Abbreviations

B/M – book-to-market ratio

CMA – conservative-minus-aggressive (investment) factor

CRSP – Center for Research in Security Prices

CSR – corporate social responsibility

ESG – environmental, social, and governance

gvkey – Compustat firm identifier (Global Company Key)

HML – high-minus-low (value) factor

HPWP – high-performance work practice

HRM – human resource management

LC, LU, LS – CRSP–Compustat link types (confirmed, unconfirmed, security-specific)

MKT – market excess-return factor

MOM – momentum factor

permno – CRSP security identifier

rcid – Revelio Labs company identifier

RMW – robust-minus-weak (profitability) factor

ROA – return on assets

S&P – Standard & Poor's

SHRM – strategic human resource management

SMB – small-minus-big (size) factor

WRDS – Wharton Research Data Services

Abstract

Does treating employees well pay off for shareholders? Previous research concluded that America's most admired employers earned a stock-market premium, but that evidence predates the decade in which employee wellbeing turned into a mainstream concern. This leaves the question of whether that premium remains after the institutionalization of employee sentiment scores online. This thesis measures employee sentiment from millions of online reviews of S&P 500 firms over 2013–2023, and asks whether companies with more satisfied workforces earn higher risk-adjusted stock returns, and whether sentiment predicts their own operating performance.

With regard to asset pricing, the pre-committed equal-weighted long–short portfolio yields no premium over the full window, and there is no evidence of one fading away within the sample. The null is not caused by the factor model: the raw spread already cannot be distinguished from zero, and risk adjustment removes only about a fifth of it. A value-weighted portfolio, closer to how Edmans (2011) built the original evidence, provides a larger estimated return but is still not statistically significant, so whether the null carries across portfolio designs remains unclear.

The results from the panel tests are more promising, even though the evidence is weak. There is a relationship between higher sentiment and faster sales growth the following year, but the estimate is only marginal and disappears when the firms that have been thinly reviewed are excluded. The link to return on assets is weaker and not cleanly forward-looking, and operating margin, the marker of pricing power, shows none at all. Worker wellbeing therefore looks like a modest business signal at best. This study does not rule out a smaller surviving residue, but it narrowly excludes a premium as large as the original one. If the signal had ever been mispriced, the pricing-in most probably took place before the sample, although the design cannot establish when it occurred, and the one break in the data runs the other way. This thesis runs on a point-in-time index sample, free of index-membership survivorship bias and of year-level look-ahead, with its specifications fixed in a dated commit before the outcome data were retrieved.

1. Introduction

A central question in both the fields of management and finance is whether spending on employee satisfaction creates firm value, and whether the stock market accurately reflects that value in its pricing. There are two competing views: the agency view argues that money spent on employee wellbeing can be a sign of managerial slack; managers may tolerate high pay or low effort of employees at the shareholders' expense (Edmans, 2011). The resource-based view, on the other hand, argues that skilled and motivated workers are a strategic resource that can bring a lasting advantage when it is valuable, rare, hard to imitate, and hard to substitute (Barney, 1991). Previous research in SHRM links such practices to firm performance (Huselid, 1995; Becker & Gerhart, 1996). If that connection holds, employee satisfaction is not only a mere measure of worker welfare; it is also a signal of workforce productivity. This thesis examines whether equity markets price that signal, using employee sentiment for S&P 500 firms over 2013–2023.

A large body of evidence suggests that workforce practices affect firm performance. Huselid (1995) found that across nearly 1,000 U.S. firms, high-performance work practices reduce employee turnover and increase productivity and financial performance. Becker and Gerhart (1996) describe HR systems as a source of competitive advantage. On the returns side, the key study is Edmans (2011), which documented the returns of a value-weighted portfolio of Fortune's "100 Best Companies to Work For". This portfolio achieved an annual four-factor alpha of about 3.5 percent from 1984 to 2009 and 2.1 percent after adjusting for industry, which Edmans sees as evidence of the market underpricing the value generated by satisfied employees. Green et al. (2019) apply this method to continuous Glassdoor reviews and find that changes in satisfaction ratings can predict future returns and earnings surprises. Research carried out by Lins et al. (2017) also shows a similar pattern: firms with high levels of social capital, partly based on their employee relations, outperformed their peers by four to seven percentage points during the 2008–2009 crisis. The signal therefore appears to carry information about both performance and returns.

Yet there are three gaps. First, Edmans's evidence only covers data up to 2009, before ESG investing became a mainstream strategy, before COVID-19 and the shift to remote work, and before the Great Resignation in 2021–2022, during which an unusual number of workers voluntarily resigned. If the premium was due to investors' failure to take into account an intangible aspect of employee satisfaction that is difficult to detect, then it should disappear

once such sentiment becomes apparent and ESG investments start to flow in. Although on the curated list itself, the premium seems to have persisted through 2020 (Boustanifar & Kang, 2022), whether the same holds for the broad large-cap market, measured from everyday reviews, has not yet been tested. Second, the “Best Companies” list is an annual, apply-to-enter designation that combines an employee survey with an assessor’s judgment (Edmans, 2011). It is not known whether the same effect is observed in continuous employee reviews rather than in this annual curated list. Third, the HR literature (Huselid, 1995; Becker & Gerhart, 1996) and the asset-pricing literature (Edmans, 2011; Green et al., 2019) rarely come together in a single study, so there is currently little recent evidence to show whether employees’ opinions predict both operating performance and stock returns in the same sample. This thesis is intended to fill the gap between management and finance.

Two questions emerge from this, one concerning management, the other finance. First, does employee sentiment extracted from aggregated employee reviews predict next-year operating performance (return on assets, operating margin, and sales growth) for S&P 500 firms over 2013–2023? Second, does the same sentiment predict risk-adjusted returns, which are returns beyond what known risk factors explain? The two questions are informative only together. If sentiment does predict both, the underpricing view holds: the market prices a real signal slowly, which therefore creates an investment opportunity. If sentiment only predicts operating performance but not returns, the signal is real, but investors have already learned to price it. If it predicts neither, the originally documented premium by Edmans may simply have belonged to its period.

This thesis’s sample covers S&P 500 constituents over 2013–2023. It is built from point-in-time membership, covering firms only in the years they were part of the index, to avoid index-look-ahead and index-membership survivorship bias. Employee sentiment is pulled from Revelio Labs review-level data via Wharton Research Data Services, aggregated from millions of individual employer reviews into a firm-year mean and lagged one year. Two designs follow, both set out in Chapter 3: a long–short portfolio sorted on sentiment and assessed against a six-factor model (Fama & French, 2015; Carhart, 1997), and panel regressions of next-year operating outcomes with firm and year fixed effects. The operating specification and the measurement checks were fixed in a dated commit before the outcome data were retrieved, to reduce the risk of changing the analysis to obtain favorable results.

Existing theory gives a firm prediction for one test and an ambiguous one for the other. With regard to operating performance, the resource-based view and the HR literature point to a positive relationship, this being due to lower turnover and higher productivity (Barney, 1991; Huselid, 1995). As for returns, the underpricing view (Edmans, 2011) forecasts a positive long-short alpha since intangible signals such as employee satisfaction are incorporated into prices slowly. Nevertheless, there is a reason to question the continued existence of the premium, since the period from 2013 to 2023 is precisely the time when ESG investing became institutionalized and when review platforms made it easier to observe sentiment. If investors have learned to correctly price in the intangible, the originally documented premium should shrink or disappear. The return prediction is therefore rather ambiguous: the premium may have survived in a weaker form, or it may have been eliminated as the signal became common knowledge across investors. The operating-performance prediction is firmer, since it is based on the real relationship between workforce and company operations rather than on how markets price it.

This thesis makes three main contributions. First, it updates Edmans's findings by testing the satisfaction premium over a period markedly different from the original sample, during which both the visibility and the valuation of employee sentiment shifted. Whether the premium survived is worth knowing in itself: if it did, it would challenge the expectation that widely known market anomalies weaken as investors learn about them (McLean & Pontiff, 2016). Second, this thesis joins two literatures that usually run in parallel: the HR research, which links firm practices to their outcomes (Huselid, 1995; Becker & Gerhart, 1996), and the asset-pricing work, which tests whether markets price them in (Edmans, 2011; Green et al., 2019). By addressing both of these questions using the same sample, this thesis not only determines whether a premium exists but also elucidates how it functions. Third, it contributes to the measurement of employee sentiment by relying on continuous, rank-and-file reviews from Revelio Labs rather than an annual, apply-to-enter list, thereby reducing concerns about curation and self-nomination that affect earlier measures. The evidence answers the two questions differently. No risk-adjusted return premium can be detected over the window, and the operating side offers only a marginal link between higher sentiment and faster sales growth the following year. The results of this study are relevant to managers when deciding whether or not investing in their workforce is worthwhile and also to investors who want to know whether sentiment contains information that is not reflected in prices.

2. Literature Review

2.1 Strategic Human Resource Management and Firm Performance

For decades, a central proposition in strategic human resource management (SHRM) has been that the way firms treat their employees shows up in their financial performance. This idea was not always broadly accepted; HR used to be regarded as a cost center which should be reduced rather than as a source of competitive advantage, and early empirical studies attracted considerable skepticism. Yet that attitude has not stood up to the evidence. There is now a substantial amount of research that links particular sets of HR practices to outcomes at the firm level, and it is this body of evidence that forms the basis of the present thesis, which treats employee sentiment as its main independent variable.

The modern literature starts with Huselid (1995), who studied nearly one thousand U.S. firms. He discovered that the adoption of what he referred to as High Performance Work Practices (HPWPs), a bundle of practices which includes selective recruiting, performance-linked compensation, employee involvement programs, and extensive training, was linked to lower employee turnover and better corporate financial performance. The estimated effects were not trivial. A one-standard-deviation increase in HPWP adoption corresponded to a 7.05 percent decrease in turnover and, per employee, to \$27,044 in additional sales, \$18,641 in additional market value, and \$3,814 in additional profits (Huselid, 1995). While earlier studies had linked single practices to performance, Huselid's was the first national-sample test of comprehensive HR systems. Its findings provided crucial evidence against the belief that HRM is merely an administrative function and supported the claim that employee-related practices can drive value creation.

Later work confirmed the pattern and refined this relationship. Combs et al.'s (2006) meta-analysis of 92 studies put the overall correlation between HPWP adoption and organizational performance at roughly 0.20 after correcting for measurement error, representing a meaningful effect by the standards of organization-level research in management. Most importantly, their analysis highlighted the value of HR systems rather than individual practices in isolation, with corrected correlations of 0.28 and 0.14, respectively. So, the evidence is in favor of bundles of practices over standalone interventions, whether through synergy or simply through their accumulation. Posthuma et al. (2013) later organized 193 peer-reviewed articles from 1992 to 2011 into a systematic taxonomy of HPWPs, building on the accumulated evidence that HPWP systems positively influence operational and financial

outcomes through intermediate mechanisms, such as job satisfaction, productivity, and turnover.

What happens inside that process has become a research question of its own. Jiang et al. (2012) meta-analyzed the mediating pathways and differentiated HR practices among three categories: skills-enhancing, motivation-enhancing, and opportunity-enhancing. Skill- and motivation-enhancing practices affected financial outcomes both directly and indirectly, whereas the effect of opportunity-enhancing practices operated entirely through mediators. The indirect effects worked through human capital and employee motivation, which subsequently shaped voluntary turnover and operational performance. Thus, the HRM-performance link is not a black box: specific practices affect specific employee-level outcomes, and these aggregate into firm-level results.

Nevertheless, it is not obvious why any of this should produce a durable competitive advantage rather than a temporary one that competitors can replicate. The standard explanation comes from Barney's (1991) resource-based view of the firm. According to this framework, a resource can sustain competitive advantage only if it meets four criteria: it must be valuable, rare, imperfectly imitable, and non-substitutable. Barney considers human capital to be one of the fundamental categories of resources within a firm and claims that socially complex phenomena such as organizational culture are examples of things that are imperfectly imitable, since they lie beyond the ability of competitors to observe and systematically reproduce (Barney, 1991). Becker and Gerhart (1996) applied this argument to HR systems, describing a well-developed system as an "invisible asset" (p. 782), a term which they have taken from Itami (1987). Since HR systems are generally marked by causal ambiguity and path dependency, it is not easy for competitors to replicate them by reverse engineering; they may not understand how the various components of the system interact, and the system itself evolves over time as a result of a series of accumulated decisions and experiences which cannot easily be reproduced. From this point of view, an effective HR system is in itself a strategic resource. However, since such systems are largely intangible and difficult for outsiders to observe, the question remains whether financial markets recognize and correctly incorporate their value into prices.

Three takeaways from this literature matter for the present thesis. First, the HRM-performance link is robust across hundreds of studies and economically meaningful in size. Second, it runs through identifiable mechanisms (skill development, motivation, retention),

which means that observable proxies for them, including employee satisfaction and sentiment, should contain information about firm performance. Third, because effective HR systems are causally ambiguous and path dependent, they represent a class of intangible firm value that markets may struggle to price. Whether financial markets actually price this intangible remains an open question.

2.2 Employee Satisfaction and Asset Pricing

For a long time, finance research has regarded the possibility that markets undervalue intangibles, and theory suggests why. Merton (1987) showed that stocks followed by fewer investors may trade at a discount, because their holders cannot spread firm-specific risk widely and must be compensated for bearing it. As a stock becomes better known, that discount shrinks. An intangible which is not closely monitored by many investors, for example employee satisfaction, is precisely the kind of item that this logic predicts the market will take a long time to price. Edmans looked at a value-weighted portfolio of companies on the “100 Best Companies to Work For in America” list, giving greater weight to larger firms, and found an annualized four-factor alpha of 3.5 percent from 1984 to 2009, of which 2.1 percent survived industry adjustment. An alpha is the return left unexplained after adjusting for the exposures that normally drive return differences; in Edmans’s case, the market, size, value, and momentum factors. A persistent positive alpha indicates that a strategy earns more than its exposure to these risk factors would justify. The result remained robust under alternative weighting schemes, with outliers removed, and against characteristics-matched benchmarks following Daniel et al. (1997), which compare each stock with peers of similar size, valuation, and recent performance. In Edmans’s extended sample through 2011 (Edmans, 2012), the outperformance returns range from 2.3 to 3.8 percent a year, depending on the benchmark used.

Edmans interpreted this as evidence that the stock market does not fully incorporate intangible human capital into stock prices, and he had three arguments. First, the positive relationship between satisfaction and shareholder returns contradicts the older managerial view, adopted by an equity analyst, whom he quoted at the start of his paper, that spending on employees is managerial slack at shareholders’ expense. Second, the outperformance persisted after Fortune began publishing the list prominently in 1998, so the mispricing could not be blamed on missing information; the satisfaction data were independently verified, public, and widely read, and markets still absorbed them slowly. Edmans (2012) suggested

that the market “uses traditional valuation methodologies devised for the 20th-century firm and based on physical assets, which cannot accommodate intangibles easily” (p. 14), and may take four to five years to fully incorporate intangible information. Third, and most damaging to a risk-based explanation, the firms on the Best Companies list posted significantly more positive earnings surprises, meaning reported profits kept beating analysts’ forecasts, and stronger price reactions when those announcements arrived. Markets were not demanding compensation for some hidden risk; they were under-anticipating these firms’ operating performance (Edmans, 2011).

Green et al. (2019) refined the design using Glassdoor.com, where review-based ratings are continuous and update constantly, unlike a once-a-year binary inclusion list. Quarterly changes in firms’ average Glassdoor ratings predicted subsequent abnormal returns, beyond what the firms’ risk exposures would explain, concentrated exactly where one would expect if the ratings carried real information: among current-employee reviews, longer reviews, and firms with high idiosyncratic volatility, the stocks whose firm-specific price swings make them risky to trade against (Green et al., 2019). The analysis of the sub-dimensions revealed the underlying mechanism: changes in the Senior Management ratings produced the strongest return predictability, with an equal-weighted long–short alpha of 0.71 percent per month, which was achieved by buying companies whose ratings had improved and selling those whose ratings had declined. Career Opportunities followed, while changes in Work/Life Balance ratings did not show any predictive power. Furthermore, rating changes were linked to simultaneous but not yet public shifts in profitability and to predicted future earnings surprises, which supports the view that this represents a fundamental information channel rather than just a reflection of investor mood. The continuous ratings and the sub-dimension breakdown are also included in the design of Chapter 3.

The fact that employee sentiment has informational value is also supported by accounting research carried out on Glassdoor.com. Hales et al. (2018) examined whether the opinions employees volunteer on the platform predict future corporate disclosures, using a sample of roughly 150,000 reviews of S&P 1500 firms. They discovered that employee evaluations, especially the explicit “six-month business outlook” question and a latent outlook factor derived from the different rating categories, could predict growth in future income statement items, temporary items such as restructuring charges, and management forecast news. These results indicate that the information generated by employees gives investors insights that go

beyond what is contained in traditional corporate disclosures (Hales et al., 2018). Furthermore, this pattern is not limited to data from the United States: Symitsi et al. (2021) find that Glassdoor ratings contribute incremental predictive power to the future financial performance of UK companies. Since employee sentiment predicts a variety of firm outcomes, not just returns, the present design requires both an asset-pricing and an operating-performance test.

The employee satisfaction premium first identified by Edmans seems to continue to exist beyond the initial sample, although with certain limitations. Boustanifar and Kang (2022) reran the Best Companies strategy through 2020 and still found a premium of about 2 to 2.7 percent a year in an equal-weighted portfolio, suggesting that the effect was not only tied to Edmans's window. Edmans et al. (2024) extended the study to thirty countries and discovered that the premium increases as a country's labor market is more flexible: it is present in flexible markets such as the United States and the United Kingdom but is less noticeable in those that are more regulated, like Germany, where firms have fewer opportunities to adjust their employment and to convert satisfied workers into better performance. Sheng (2025) also makes a distinction between employees' current feelings (backward-looking satisfaction) and their future expectations about the company (future-looking expectations): the latter are better at predicting returns than the former, and sophisticated investors seem to take account of them.

An alternative explanation is that employee-related characteristics might be priced as risk rather than mispriced. Eisfeldt and Papanikolaou (2013) develop a model in which organization capital is a production factor embodied in a firm's key talent. Because this capital lives partly in people, both shareholders and key employees hold claims on its cash flows, and their split depends on the employees' outside options, which makes high-organization-capital firms riskier and, in consequence, gives them higher expected returns. Empirically, such firms obtain average returns which are 4.6 percentage points higher than firms that have little organization capital (Eisfeldt & Papanikolaou, 2013). If, by hypothesis, firms with high sentiment are rich in organization capital, then they might be systematically larger and more profitable, and the issue then becomes whether the factor models correctly price this kind of exposure.

A parallel literature on corporate culture provides insight into which part of the employment experience is most informative: the signal comes from management quality rather than perks.

Guiso et al. (2015) find that values firms advertise publicly are unrelated to actual performance, whereas the dimension employees actually perceive, senior managers' trustworthiness and integrity, does predict it. Graham et al. (2022) reach a similar conclusion by means of a survey of more than 1,300 executives, who regard corporate culture as one of the main factors influencing firm value and credit this effect to leadership quality. Gorton et al. (2022) review this evidence. This thesis also returns to this distinction in its own findings, looking at whether the different sub-dimensions of sentiment differ in their ability to predict firm performance.

Faleye and Trahan (2011), in their study of companies that Fortune had classified as employee-oriented, discovered that labor-friendly firms performed better than their counterparts on both stock market and operational indicators over a number of years, the most significant effects being seen in those firms which relied most heavily on skilled human capital. In contrast, earlier work pointed the other way: Faleye et al. (2006) found that firms with employees who held formal governance power through large equity stakes were associated with depressed sales growth, investment, and value. Therefore, informal employee engagement seems to help, while formal control seems to hurt. Lins et al. (2017) further added evidence from the 2008–2009 financial crisis, during which high-CSR firms, with employee relations a major component of the score, earned stock returns four to seven percentage points above low-CSR firms. They argue that social capital pays off when trust becomes scarce, which makes the pricing of employee-related factors potentially state-contingent.

Taken together, the literature points to a consistent pattern. During the period 1984 to 2009, employee satisfaction was found to predict risk-adjusted returns, the most well-supported explanation being mispriced intangible human capital. Beyond stock returns, sentiment has also been linked to operating performance and corporate disclosure, and it works through particular sub-dimensions. Among these, evaluations of senior leadership show the greatest predictive power, something that is in line with the resource-based view of management quality as a strategic resource. Employee relations is also one of the ESG dimensions that accounting research finds financially material in many industries (Khan et al., 2016), which is part of why it might be expected to be priced where less material factors are not. All of this, however, was largely based on data collected pre-2010, before Glassdoor made employee

sentiment publicly observable and before ESG-oriented institutional investment went mainstream. Whether those findings survive the decade that followed is another question.

2.3 The Modern Era: Anomaly Decay, ESG Pricing, and Crisis Evidence

Several developments during the past decade, which is this thesis's examination period, give reason to think the satisfaction-return relationship may have changed. The question relates to three areas of the literature: the post-publication decline of return anomalies, that is, patterns which predict returns better than risk by itself would indicate; evidence concerning the effect of ESG-focused investing on expected returns; and evidence from times of crisis as to whether certain characteristics tend to increase in value under particular market conditions. The underlying idea is that some firm characteristics pay off mainly during periods of economic stress.

The reason for expecting the satisfaction premium to weaken is most clearly given by the literature on anomaly decay. McLean and Pontiff (2016) examined the out-of-sample and post-publication performance of 97 variables, previously shown to predict cross-sectional stock returns, meaning variables that could identify which stocks would do better in advance. Portfolio returns fell 26 percent out-of-sample and 58 percent after journal publication. They interpret the out-of-sample decline as an upper bound on data-mining bias, leaving an additional publication-specific effect of about 32 percentage points that they attribute to investors learning of the mispricing and trading it away (McLean & Pontiff, 2016). More broadly, anomaly returns are sensitive to methodological choices and, for some predictors, are concentrated in the smallest stocks (Fama & French, 2008). The decline was larger for predictors that had higher in-sample returns and smaller for those that were concentrated in hard-to-arbitrage stocks, where taking advantage of the mispricing is costly. Both findings are consistent with a correction of mispricing rather than with rational risk compensation. The logic applies directly to the satisfaction premium: Edmans (2011) appeared in a leading journal, sentiment then became broadly observable through Glassdoor, and if the original premium ever reflected mispricing, it should have shrunk over the studied period.

The clearest example of how investors' learning can change a return pattern can be found in the research concerning corporate governance. An index, built by Gompers et al. (2003), that measures governance provisions limiting shareholder rights indicated that firms with the fewest such provisions, meaning firms with stronger shareholder-friendly governance, generated substantially higher risk-adjusted returns through the 1990s. Similar to Edmans's

findings, this premium was interpreted as the market underpricing an intangible. Bebczuk et al. (2013) later showed the premium faded to nothing over the following decade, as corporate governance became a widely recognized investment concern. This kind of pattern is characteristic of a mispricing being eliminated once investors have had enough time to learn to look for it. Core et al. (2006) had already suggested a more cautious interpretation at an earlier stage, observing that the governance premium was not accompanied by the negative earnings surprises that one would expect if the market had not recognized the true value of governance. The analogy with employee satisfaction follows the same pattern: an intangible that is initially ignored gains an abnormal return early on, and the return should disappear as the signal becomes generally known. Core et al.'s caution carries over in part, but not fully: the satisfaction case rests on the positive earnings surprises documented in Section 2.2 (Edmans, 2011), which is precisely the evidence the governance premium lacked.

There is another mechanism that works as a result of investor demand. According to their ESG-efficient frontier framework, Pedersen et al. (2021) show theoretically that when certain investors prefer companies with good ESG characteristics, the intensity and extent of these preferences influence equilibrium expected returns. Pástor et al. (2022) examine this idea using U.S. data and find that green stocks, which are highly environmentally rated, outperform brown stocks, which score lower on the environmental ratings, by a cumulative 174 percent over 2012–2020. This result was a let-down for green investors since the outperformance was caused by an unexpected rise in environmental concerns, not because of higher expected future returns. After accounting for actual returns and adjusting for changes in climate concern and earnings shocks, they found that the expected return on green stocks was actually below that on brown stocks, precisely as the theory states when investors drive up the prices of the assets they prefer. The demand channel is also supported by empirical evidence. Using Morningstar's introduction of sustainability ratings as a natural experiment, Hartzmark and Sussman (2019) show that investors moved large amounts of capital into funds newly labeled as high-sustainable and away from those labeled low, direct evidence that a non-financial label can move investor demand by itself. Some of that demand is a conscious trade of return for values: Riedl and Smeets (2017) find that many socially responsible investors knowingly accept lower returns, a preference Bénabou and Tirole (2010) incorporate into their economic model of corporate social responsibility. If employee-friendly firms became favored assets in the same way over 2013–2023, the same logic predicts their expected returns fell.

The demand mechanism actually existed before the ESG era; Hong and Kacperczyk (2009) showed its counterpart in the case of “sin” stocks, that is, publicly traded companies in the alcohol, tobacco, and gaming sectors that were avoided by institutions bound by social norms such as pension funds. These sin stocks received less analyst coverage and earned higher returns than otherwise comparable stocks, a price effect the authors put at 15 to 20 percent (Hong & Kacperczyk, 2009). What is important is the symmetry between the two situations. Companies which investors steer clear of for non-financial reasons trade at a discount and deliver higher returns; the premium-bearing case represents the reverse of this situation. If during the sample period a sufficient number of investors had come to prefer companies that are friendly to their employees, either because of explicit ESG requirements or simply because data on employee sentiment had become available, the expected returns of such firms should have fallen.

Lins et al. (2017) provide a more complex view of the premium’s expected decline. As Section 2.2 noted, the outperformance they document is concentrated in the 2008–2009 crisis, a period when trust in both corporations and the markets underwent a severe setback. The premium associated with employee-related factors therefore appears not to accumulate gradually but to arrive under particular economic conditions. The time frame examined in this thesis also includes the 2020 shock caused by COVID-19, a disruption that, according to their argument, should have resulted in a reward for employee-related factors; it also brought about major changes in the labor market.

From 2020 to 2022, the pandemic had a major impact on working arrangements and led to what was termed the “Great Resignation,” a sharp increase in people voluntarily quitting, which raised the U.S. quit rate to a record in late 2021 and kept it at a level close to that figure for a large part of 2022 (U.S. Bureau of Labor Statistics, 2026). Remote work spread, wage growth concentrated among lower-paid workers, and job mobility jumped. How a renegotiation of the employment relationship on that scale changes the sentiment-performance link is an open question, and a panel spanning both the pre- and post-2020 periods lets this thesis examine it.

The post-publication decay of anomalies and the ESG-driven demand that increases the attractiveness of employee-friendly firms occurred alongside the labor-market disruptions that followed 2020. Taken together, it appears that the employee-satisfaction premium which

Edmans found for the period 1984 to 2009 may have diminished or vanished in the years that followed.

2.4 Synthesis and Predictions

The literature reviewed above motivates two empirical questions, with different expectations behind each one.

The first concerns operating performance. As mentioned in Section 2.1, the ways in which firms manage their employees affect financial results via three channels: productivity, employee retention, and skill development. Section 2.2 added that review-based sentiment contains forward-looking information about fundamentals. Green et al. (2019) found that changes in sentiment were linked to later profitability and to earnings surprises, while Hales et al. (2018) found that employee assessments predicted future income statement items and disclosure outcomes. In view of both bodies of research, the idea that sentiment should be able to predict subsequent operating performance is quite strong. Most importantly, this expectation does not rely on how the market prices the signal; instead, it is based on the actual economic relationship between satisfied employees and the firm's output, a relationship that is directly tested by the panel regressions of operating performance on lagged sentiment (H2b).

It is more difficult to predict stock returns since the literature presents conflicting views. As has already been shown, Edmans (2011, 2012) found a risk-adjusted premium associated with employee satisfaction between 1984 and 2009 and interpreted this as evidence of mispricing in intangible human capital; replications up to 2020 (Boustanifar & Kang, 2022) and across countries (Edmans et al., 2024) indicate that this phenomenon was not confined to the original sample period, and Eisfeldt and Papanikolaou (2013) have provided a theoretical basis for organization capital being priced. However, against this stand the three forces from Section 2.3, each specific to the reviewed period. The first is that published anomalies tend to disappear after investors become aware of them and start to trade on them (McLean & Pontiff, 2016). The second is that, over these same years, ESG-oriented investment became institutionalized, and the logic of Pedersen et al. (2021) and Pástor et al. (2022), with the sin-stock evidence of Hong and Kacperczyk (2009) as its counterpart, states that such demand causes favored firms to be bid up and reduces their expected returns. The third point is that the labor market after 2020 was very different from the one in which the original evidence was obtained. The asset-pricing prediction is therefore genuinely uncertain: the premium

could have survived, but it could just as plausibly have weakened or disappeared. This is tested using portfolio sorts and a factor-model alpha regression (H1 and H2a).

No matter what the results of the interpretation may be, there is one caution to bear in mind. Daniel and Titman (1997) found evidence that the size and value premia trace to firm characteristics rather than to the covariance structure of returns: in their 1973–1993 tests, small stocks and high book-to-market stocks, priced low relative to their accounting value, earned high average returns whether or not their returns moved together with other such stocks, and the authors did not rule out a factor-model explanation. The same point applies in this case. The premium associated with high-sentiment firms could be due to a distinct sentiment signal, or it might reflect exposures that are already priced. Size is a likely suspect, since both the coverage of sentiment data and the extent to which firms are regarded as ‘best employers’ increase as firm size increases. Separating a genuine sentiment effect from masked familiar characteristics is essential. The factor model introduced in Chapter 3 controls for compensation associated with known factor exposures, and whether the characteristics themselves retain any residual is a distinction Section 5.3 returns to.

This sets the terms for the empirical work, and it can be stated as three hypotheses. **H1**: a portfolio that is long the highest-sentiment quintile of index firms and short the lowest earns a positive average raw return. **H2a**: the same portfolio earns a positive alpha after controlling for the six-factor model. **H2b**: higher employee sentiment predicts higher operating performance the following year, in sales growth, return on assets, and operating margin. H1 and H2a represent the raw and risk-adjusted versions of one return question; H2b is the operating-performance question. There is a strong prior expectation for H2b. On the other hand, it remains an open question whether sentiment continues to predict risk-adjusted returns (H1 and H2a), and any return relationship must be evaluated after adjusting for characteristics already priced into the market. The research gap is therefore just empirical: no previous study has tested the Edmans framework using continuous, review-based sentiment over the period 2013 to 2023, which is the timeframe during which employee-sentiment data became widely available, and ESG investing became mainstream. Chapter 3 presents the data and methodology, with the results given in Chapter 4.

3. Methodology

The panel, which looks at the S&P 500 constituents from 2013 to 2023, connects employee sentiment with firm performance, and for this purpose five different data sources are brought together, each one addressing a separate aspect of the question.

3.1 Sample and Data Sources

Sample period. The sample period extends from 2013 to 2023, with the year 2012 being used merely for the purpose of establishing the initial lagged sentiment figure. This period was selected for two reasons. The first is that Revelio’s coverage of reviews for companies in the S&P 500 becomes sufficient from 2012 onwards, since 82.9 percent of the firms then have at least one review; using earlier years would have meant trading sample size for noise. The second is that the period includes several significant events: the widespread adoption of review platforms, the growing institutional interest in ESG investing, and the labor shocks caused by COVID-19 and the Great Resignation, any of which might have affected the relationship between satisfaction and returns.

S&P 500 universe. Year-end membership for 2012 through 2024 comes from the public fja05680/sp500 repository (Aultman, 2026), which maintains a point-in-time membership by extending a published historical constituent list with documented index changes. This source was necessary because WU Vienna’s WRDS subscription excludes CRSP’s historical index-membership tables, and S&P Capital IQ does not expose point-in-time membership with WU’s subscription tier. The reconstructed panel was verified against publicly available records of membership events, such as the addition of Tesla in December 2020, Twitter’s entry in 2018 and its exit following the 2022 acquisition, the change from Facebook to Meta and General Electric’s ongoing membership during the period in question. All these events were accurately reproduced, and the cross-sections average about 503 tickers per year, which is in line with the size of the index. The panel holds 6,535 firm-year observations across 751 tickers and guards against both index-driven survivorship and look-ahead bias: firms that later left the index are kept for the years they belonged to it, and firms that later joined enter only from the year they were added.

Employee sentiment. Sentiment is built from review-level records in Revelio Labs’ individual-reviews table via WRDS, which holds about 43.5 million reviews worldwide, 4.30 million of them for universe firms over 2010–2024. Each review gives an overall one-to-five rating and may add five sub-ratings: work–life balance, compensation and benefits, career

opportunities, culture and values, and senior leadership. By working at the review level rather than relying on pre-aggregated data, the methodology has more flexibility: it enables filtering by employment status, retains the sub-ratings, and allows for aggregation to firm-year averages together with the number of reviews. Green et al. (2019), who have the most similar approach in design, calculate sentiment from Glassdoor ratings, whereas the sentiment used in this thesis comes from the aggregated employer reviews provided by Revelio Labs. The comparison therefore rests on the two measuring the same construct, not on the reviews coming from the same source.

Other sources. Monthly stock returns come from CRSP (crsp.msf), restricted to primary common-equity listings through the CRSP-Compustat link table; market equity uses the absolute value of month-end price times shares, following CRSP's negative-price convention for bid-ask midpoints. The annual accounting data are taken from Compustat North America (comp.funda) under the standard industrial-format filters. The monthly factor returns are sourced from the Kenneth R. French Data Library. The variables drawn from these sources are defined in Section 3.3.

The analytical sample. The merged panels span 2013–2024; 2024 stays outside the headline window and enters only the raw-return robustness check described in Section 4.2. In both panels, membership is based on point-in-time criteria. For the period 2013 to 2023, which is the one during which all of the headline tests are carried out, the asset-pricing sample includes 59,609 firm-months from 626 firms and the operating-performance sample includes 4,962 firm-years from 625 firms.

3.2 Sample Construction and Identifier Resolution

Four different firm identifiers are used by the five sources: tickers for the S&P 500 universe, gvkey from Compustat, permno from CRSP, and rcid from Revelio. Since none of these identifiers can be properly matched with one another when renamings, reused tickers, and restructurings are taken into account, each link is established explicitly and the coverage associated with it is recorded.

From tickers to gvkey. Tickers are matched to gvkey through Compustat's historical security-identifier table (comp.sec_idhist), which records ticker assignments over time; tables that hold only current tickers fail for renamed firms. Each firm-year is matched to the gvkey whose spell covers its snapshot date. The mapping is kept at the (ticker, year) level because

eight tickers are associated with two gvkeys after corporate successions. A separate eight tickers required manual matching: four had no usable spell at all, and the rest collided with stale spells of unrelated firms under the source’s forward-corrected tickers, a problem found by auditing all 751 tickers individually. The mapping covers every firm-year and yields 717 unique gvkeys.

From gvkey to permno. CRSP returns are linked by permno through the CRSP-Compustat linking table, restricted to primary links and link types LU, LC, and LS. The LS type is kept because two firms, Eaton and Dayforce, carry their only primary link as LS, and an LU/LC-only filter would zero out their returns (LS marks a link valid for that security only; primacy itself is enforced separately, with the link-primacy flag restricted to primary listings throughout). All 717 gvkeys obtain a link, and 99.8 percent of firm-years have return data.

From gvkey to rcid. Revelio reviews are linked by rcid through Revelio’s company-mapping table; 708 of the 717 gvkeys map one-to-one. United Airlines is among them only after a manual repair, because its reviews live under a subsidiary identifier. The remaining nine firms have no Revelio coverage, mostly members acquired or bankrupt before 2016, plus two 2024 spinoffs and one recent index entrant; Section 3.6 discusses the implication.

Filtering and aggregation. Among the 708 firms included in the study, reviews are retained only if the reviewer’s status is either regular or not stated. The explicitly non-core status categories, such as interns, contractors, freelancers, and part-time, temporary and seasonal workers are eliminated, thereby keeping 3,285,976 of the 4,299,282 reviews, which is 76.4 percent. Each review’s overall rating as well as the sub-ratings it has completed are averaged within each firm-year, and the mean is stored together with the number of reviews (*n_reviews*) for use in the precision-weighted check in Section 3.5.

Assembling the panels. The firm-month asset-pricing panel links membership to CRSP returns, attaches the prior year’s sentiment as a constant over the outcome year, and merges the factor returns for both year and month, since CRSP records the month-end on the last trading day and the factor library on the last calendar day. Membership acts as a row filter, not just a firm list: firm-months outside a firm’s own membership years are excluded. On the 2013–2023 headline window the panel holds 59,609 firm-months across 626 firms. The firm-year operating-performance panel joins membership to Compustat fundamentals under the same one-year lag and the same membership rule, and on the same window holds 4,962 firm-

years across 625 firms. The numbers agree with the universe as follows: the 751 tickers correspond to 695 different firms that have membership spanning 2013 to 2023; 48 of these never provide a reviewed firm-year and an additional 21 have no lagged score within their membership years, which accounts for the 626 firms in the return panel. The operating panel loses one more firm, FMC Technologies, since the only firm-year it has that includes both membership and a lagged score, namely 2016, has no Compustat record using its pre-merger identifier. The coverage of sentiment increases from 82.9 percent of firms in 2012 to more than 95 percent from 2021, while the coverage of returns and fundamentals stays above 99 percent throughout.

3.3 Variable Definitions

Employee sentiment. The explanatory variable, employee sentiment, is the firm's average overall rating on a one-to-five scale for each firm-year, lagged by one year. For instance, the reviews posted by a firm in 2018 are linked to its results in 2019. This lag serves two purposes. First, since the return tests are carried out on a calendar-month basis, it ensures that sentiment is assessed before the outcomes it is intended to predict. For accounting outcomes, the timing is less exact, where a fiscal year ending mid-year can overlap the review year by up to six months. Second, it also mirrors Edmans (2011), where satisfaction is measured before returns. The five sub-ratings are built the same way for the tests in Section 3.5, and `n_reviews` records the number of reviews that stand behind each firm-year average.

Operating outcomes. The operating outcomes employed in this thesis are three accounting measures. Return on assets (net income over total assets) captures profitability. Operating margin (operating income before depreciation over revenue) captures how much of each sales dollar becomes profit. Sales growth (the year-over-year percentage change in revenue) captures expansion. Each of these measures is winsorized at both the 1st and 99th percentiles, meaning that the most extreme 1 percent of values at each end are cut off in order to prevent a small number of outliers from influencing the estimates. The winsorization is applied at the estimation stage. Firm-years missing an outcome are dropped before winsorizing, not kept at the clipped bound, and the stored panels keep the original raw values.

Controls and returns. The panel regressions add two control variables: the logarithm of total assets, a standard size measure, and leverage (total liabilities over total assets, winsorized just like the outcomes). These controls are included because larger and more indebted firms differ in profitability and growth. On the asset-pricing side, monthly total returns are used. Risk

adjustment is carried out using the factor model, and as a result, no firm-level controls are included directly.

Risk factors. The factor model includes the five Fama and French (2015) factors (market, size, value, profitability, and investment) plus Carhart's (1997) momentum factor, all from the French Data Library. The inclusion is crucial to measure how much of a portfolio's return can be explained by exposure to well-known investment styles. The remainder, the alpha, represents the performance beyond the known factors. Appendix A provides the full variable definitions and source fields.

3.4 Empirical Specifications

This thesis uses two families of tests developed in Section 2.4: whether the financial markets price employee sentiment and whether sentiment predicts actual operating performance. Examining both relationships is important, since sentiment could predict operating results but not earn any excess return, which is consistent with the outcome several forces in Section 2.3 would produce.

Portfolio sorts (H1). Each year, every firm with a lagged sentiment score is sorted into five equal groups, from Q1 (lowest sentiment) to Q5 (highest sentiment). Each quintile's equal-weighted monthly return is calculated over twelve months, with portfolio weights recalculated each month. The portfolios are reconstructed every January. Equal weighting gives every firm equal importance and prevents a handful of large-caps from dominating a quintile. This deliberate choice was fixed before any results were seen, and a post-hoc check with a value-weighted variant is reported in Section 4.3. The main interest is the spread portfolio, long Q5 and short Q1. This strategy buys the highest-sentiment fifth and sells the lowest, so its return is positive when high-sentiment firms outperform low-sentiment ones. The absence of a return-relevant signal should result in the spread hovering near zero. Its average monthly return is assessed with a simple one-sample t-test; the Newey–West correction described below applies to the factor regressions, not to this raw spread.

Alpha regression (H2a). A raw spread can mislead, because high-sentiment firms might simply be large or profitable firms. The second test therefore regresses the monthly spread on the six factors mentioned in Section 3.3. The window contains 132 monthly observations. Since the long–short portfolio is a zero-cost strategy, its return is already measured as an excess return, so the risk-free rate is not subtracted again. Standard errors follow Newey and

West (1987) with three lags to account for autocorrelation and changing volatility. These three lags are a conventional choice for monthly portfolio returns, which generally show limited serial correlation even when holdings persist. The results are robust to alternative specifications such as the six- and twelve-lag variants, reported in Section 4.3.

Panel regressions (H2b). The operating-performance tests regress each year- t outcome on year $t-1$ sentiment, alongside of log assets and leverage control variables. The models include firm and year fixed effects and standard errors clustered at firm level. The fixed effects determine how the results should be interpreted. Firm fixed effects compare each company only with itself over time, meaning that the estimate cannot come from stable differences between, for example, software firms and utilities. Year fixed effects capture shocks common to all firms in a given year, including the jump in average ratings that enters the 2021 outcome year. The remaining variation captures whether a firm whose sentiment rises above its own norm posts above-norm results the next year. Clustering standard errors lets each firm's errors correlate across years rather than treating every firm-year as independent.

3.5 Pre-Committed Robustness Specifications

Pre-commitment helps prevent specifications being changed after results have been obtained to shift the research in a favorable direction, and its scope should be therefore cleanly defined. The aggregation rule, equal weighting approach, the unweighted headline panel, the review-count threshold, and the precision-weighting method were specified in the pipeline and recorded in a dated git commit (9ed24f0, May 14, 2026) the day before the return and accounting data were retrieved. Therefore, the results could not have influenced those specification choices. Although the falsification and sub-rating designs mentioned below were prepared as part of the same robustness framework, they were entered into version control together with their first results a week later, and this thesis therefore refers to them as pre-specified rather than pre-committed. The protection covers the specifications but not the sample construction. Three data-production errors were discovered following the initial results: difficulties with identifier mapping, membership filtering, and lag construction in three places. All three have now been corrected in accordance with a materiality rule which had been defined in advance. Under that rule, the corrected figures are adopted directly only if the correction moves no headline t -statistic across the 1.96 or 1.65 threshold and changes no headline coefficient by more than about 10 percent. The correction relating to membership did violate that boundary and was reported to the supervisor before being implemented; the

other two remained within the boundary. All the corrections are disclosed in Section 3.6. The complete pipeline and its commit history are submitted as a supplementary file with this thesis.

Review-count threshold. The panel regressions are carried out using only those firm-years that have at least 30 reviews. This exercise seeks to determine whether the main findings rely on the less reliable sentiment estimates from the early years, which are thinly covered.

Precision weighting. As an alternative method, all the firm-years are retained but are given a weight equal to the square root of the number of reviews they have, so that more accurately measured observations carry greater weight. The review threshold and the precision-weighting methods deal with the same kind of noise from two different angles: the threshold eliminates the noisier observations while weighting reduces their impact.

Sub-rating tests. The panel tests for return on assets and sales growth are repeated using each of the five sub-ratings instead of the overall sentiment score. The tests examine the dimension along which the signal works; Figure 5 in Section 4.5 also includes the operating margin, even though Table 5 does not, thus giving a full picture. As the sub-ratings are highly correlated and since running multiple tests raises the likelihood of coming across significant results by chance (Harvey et al., 2016), the analysis concentrates on broader patterns rather than interpreting the individual marginal results as a precise ranking.

Falsification. If sentiment merely co-moves with and reflects persistently good performance instead of preceding it, then year $t-1$ sentiment should also ‘predict’ year $t-2$ outcomes just about as well as year- t ones. The falsification test examines this backward-looking relationship. A truly forward-looking signal should not show any predictive power in this backward test, and any such backward association is regarded as a limitation.

Subperiods. The sample is also divided at January 2020. The break date was imposed on the basis of the pandemic and the labor-market disruption that followed, rather than estimated from the data. Since it was in part motivated by the raw return series, it is treated as providing a background context for the full-sample estimates and not as a pre-committed test. A formal assessment of stability also includes two post-hoc tests: a Chow (1960) test at the break point fixed at January 2020 and an Andrews (1993) sup-Wald test which examines possible unknown break dates. Neither of these tests was specified in advance, so they are presented as exploratory evidence and not as pre-committed hypothesis tests.

3.6 Methods Caveats and Disclosures

Identifier handling. Since companies are listed in the universe using their most recent tickers, linking them to historical databases is prone to error. The audit noted in Section 3.2 corrected twelve mis-assignments: the four stale-identifier collisions described there, plus eight lineage tickers that had been collapsed onto a single entity even though they represented separate firms. Section 3.2's eight manual assignments refer to a different count: the four collisions plus the four tickers with no usable spell. Therefore, the two counts overlap but do not represent the same corrections. The corrected mapping and a before-and-after comparison of every headline estimate are preserved in the version-controlled pipeline, and no qualitative conclusion changed.

Membership filtering. An audit of the assembled panels found that an earlier version treated index membership as a firm list rather than a row filter, which admitted roughly a fifth of firm-observations from years outside index membership, with over half of them representing periods before later index entrants were added. The merge was corrected and the pipeline re-run before the results were finalized. As a result of the correction, the operating estimates were lowered and an apparent pre-2020 return premium was eliminated, a point which is discussed further in Section 5.4. The corrected code and the before-and-after comparison are stored in the version-controlled pipeline, as with the identifier audit.

Lag construction. A later audit found that three calculations took prior-year values by row position rather than by fiscal year, which could create problems when firms had gaps in their membership history: the falsification test, the persistence figure, and the sales-growth variable itself. In each of these cases, the errors were fixed by matching on the fiscal year. In the sales-growth case, the defect fabricated fourteen growth values, none of which entered the analytical sample, because the membership filter had already excluded them. No headline estimate changed, and the falsification statistics in Section 4.5 are the corrected values.

Membership granularity. The universe is composed of snapshots of the index taken at the end of each year; as a result, a firm is treated as a member for the entire calendar year if it joins during the year, and a firm that leaves during the year is dropped from it entirely. Consequently, less than two percent of firm-months occur before the firm's actual entry into the index. When membership is rebuilt on a monthly basis, the full-window alpha rises from 1.43 to about 1.5 percent per year, and all the conclusions remain the same, which is why the convention is explained here rather than being corrected.

Sentiment coverage. Fifty-three universe firms never contribute a reviewed firm-year, nine of which cannot be matched to Revelio at all. Among the 695 firms with S&P 500 membership during the audited period, 48 of the 53 fall among the firms inside the headline window, matching the count in Section 3.2. Because the missing firms are mostly ones that were acquired, merged, or went bankrupt during the window, the tilt runs toward survivors. If declining sentiment is associated with future declining performance, excluding them most likely weakens (biases downward) the sentiment-performance link rather than creating a false-positive one. However, this cannot be checked directly, because the missing firms left no sentiment to check.

Reviewer status. The sentiment means keep reviews whose employment status is recorded as regular or left blank, and exclude the explicitly non-core status categories (interns, contractors, freelancers, and part-time, temporary, and seasonal staff). Blank status is largely an early-sample artifact. It falls from about two-fifths of kept reviews in 2012 to under one percent from 2020 onward as the platform's field capture matured, so it reflects the timing of a review more than the kind of worker who wrote it. A filter that only includes regular employees would mainly reduce the coverage during the earlier years. Any non-core reviews that still manage to be included will introduce measurement error. If this error has no relation to the firm's performance, it will cause the estimates to be biased towards zero, but if the error varies over time or with review volume it could have other effects. This is one of the reasons why the concentration of the operating result before 2020 should be treated with caution in Section 5.4.

Measurement precision. Review volume rises considerably during the period under review: in 2012 the median firm-year is based on about 40 reviews but in 2021 it is based on more than 300. Consequently, the early averages are noisier, and noise in an explanatory variable reduces the estimates, a problem that is tackled by the review-count threshold and the precision weighting.

Crowdsourced data. There are three limitations relating to crowdsourced employee reviews which no subsequent step can overcome, and all of them are mentioned below. The first is that the people who write the reviews select for themselves and therefore do not represent a random sample of the entire workforce, a fact that no reweighting can completely correct. Second, companies might also try to affect the measure: on consumer platforms, businesses can encourage and even plant positive reviews (Luca & Zervas, 2016; Mayzlin et al., 2014),

and similar incentives may be available on employer review websites. If such profile management differs across firms, it may introduce systematic error rather than averaging out. Third, the vendor revises history: re-pulling the same firm-years four weeks later shifted 0.7 percent of them slightly.

Delisting returns. CRSP's monthly return file does not include the return in a stock's delisting month, and the pipeline does not add these returns back. Because delisting months are often associated with large negative returns, omitting them makes the affected stocks appear to perform better than they actually did. These firms are more likely to appear in the low-sentiment portfolio. As a result, the short leg of the portfolio seems to have performed better than it truly did, therefore reducing the measured spread. The effect is expected to be small because firms typically leave the S&P 500 before they delist, limiting the number of affected observations. The issue is therefore disclosed as a limitation rather than corrected.

Timing granularity. Employee sentiment is measured annually, while returns are monthly. So, within an outcome year the signal is stale by construction, up to 23 months between the earliest contributing review and the last matched return month. This timing limitation may weaken the ability to detect a return signal, a caveat to keep in view for the asset-pricing results. The panels include 2024, although this thesis focuses on the 2013–2023 headline window. Section 4.2 notes the raw-return check including 2024.

4. Results

This chapter presents the empirical findings. It starts by outlining the sample and explaining how the variables behave (Section 4.1). It then looks at both research questions, beginning with the return tests, since their results set the tone for the rest: whether employee sentiment predicts stock returns (Sections 4.2 and 4.3, H1 and H2a) and whether it can predict the firm's own operating results (Section 4.4, H2b). The findings are followed by a series of robustness checks (Section 4.5). The results are all based on the period from 2013 to 2023; data from 2024 is included only in those cases where it is mentioned. The text includes tables and figures for visual illustration.

4.1 Sample Characteristics

Table 1 gives an overview of the sample. The average firm-year has an employee-sentiment score of 3.50 on the one-to-five scale, and the middle 50 percent of the firm-years are situated between approximately 3.2 and 3.8. The distribution is only slightly left-skewed, the majority

of the observations being found in the upper half of the scale. This shows that it is uncommon for workers to be genuinely unhappy and that the variations in sentiment among firms are relatively small, the standard deviation being about 0.5. The number of reviews behind each score varies much more. The median firm-year rests on 124 reviews, but the mean is about 440 and the maximum approaches 39,000, a long right tail driven by the largest firms with the highest number of employee reviews. That unevenness is the reason why the robustness checks later weigh firms by how many reviews stand behind their score. The accounting outcomes are in line with what is expected of large publicly listed firms: the return on assets is 6.5 percent, the operating margin 25 percent, and annual sales growth is 8 percent. The wide ranges for margin and growth are due to a small number of extreme firm-years which are cut off by the winsorization in the regressions.

Table 1. *Raw values of the 2013–2023 analytical sample.*

Variable	N	Mean	Median	SD	Min	Max
Employee sentiment (lagged)	4,962	3.50	3.52	0.48	1.00	5.00
Number of reviews (lagged)	4,962	443	124	1,446	1	38,921
Return on assets	4,962	0.065	0.055	0.075	−0.85	0.51
Operating margin	4,731	0.251	0.230	0.175	−2.87	0.82
Sales growth	4,960	0.082	0.053	0.396	−0.84	21.99
Log total assets	4,962	10.08	9.93	1.31	6.86	15.17
Leverage	4,956	0.654	0.643	0.240	0.03	3.61

Over the period in question, sentiment shows two distinct patterns. Firstly, it rises gradually and then makes a sudden increase; the average score (dated by outcome year, so the 2021

figure comes from reviews posted in 2020) goes up slowly from about 3.26 in 2013 to 3.49 in 2020, before jumping sharply to around 3.73 in 2021 and then staying at that level. This jump lines up with the pandemic-era labor market, though it might instead reflect a change in reviewers' behavior or in the vendor's data collection methods. However, whatever the reason, the year fixed effects in the panel analyses take into account any change that affects all firms in a given year, so the jump has no effect on the results. Secondly, sentiment is relatively persistent within firms. The year-to-year correlation is around 0.6, meaning that a firm that scores well tends to keep scoring well. Its variation splits almost evenly between differences across firms and movement within firms over time, an analysis-of-variance share of roughly half each, and that within-firm movement is what the panel tests exploit by comparing each firm against its own usual level.

Table 2 shows that sentiment is not just a relabeling of something already known to move returns. Its strongest correlation with any firm characteristic is with book-to-market, and even that is only -0.12 ; with size the correlation is 0.10 , with profitability 0.10 , and with leverage and sales growth it is below 0.08 in absolute value. Sentiment is therefore not a simple relabeling of size or value, though the modest size link is worth holding onto: as Section 4.3 shows, even a correlation of 0.10 becomes a real factor exposure once firms are sorted into extreme groups.

Table 2. *Correlations of employee sentiment with firm characteristics.*

	Sentiment	Size	B/M	ROA	Margin	Leverage	Sales growth
Sentiment	—	0.10	-0.12	0.10	0.05	-0.02	0.07

4.2 Do High-Sentiment Firms Earn Higher Returns? (H1)

The first asset-pricing test sorts firms each year into five groups by their prior-year sentiment and compares the groups' stock returns over the following twelve months. The sort behaves exactly as intended. Average sentiment climbs steadily from 2.88 in the lowest group (Q1) to 4.06 in the highest (Q5), so the five groups really do run from the least to the most satisfied workforces, with no bunching in the middle. Figure 1 plots each group's average monthly return. The bars are close to flat. There is no upward march from Q1 to Q5 of the kind a return premium would create. The lowest sentiment group is slightly lower than the other

four, although the difference between all five groups is practically zero.

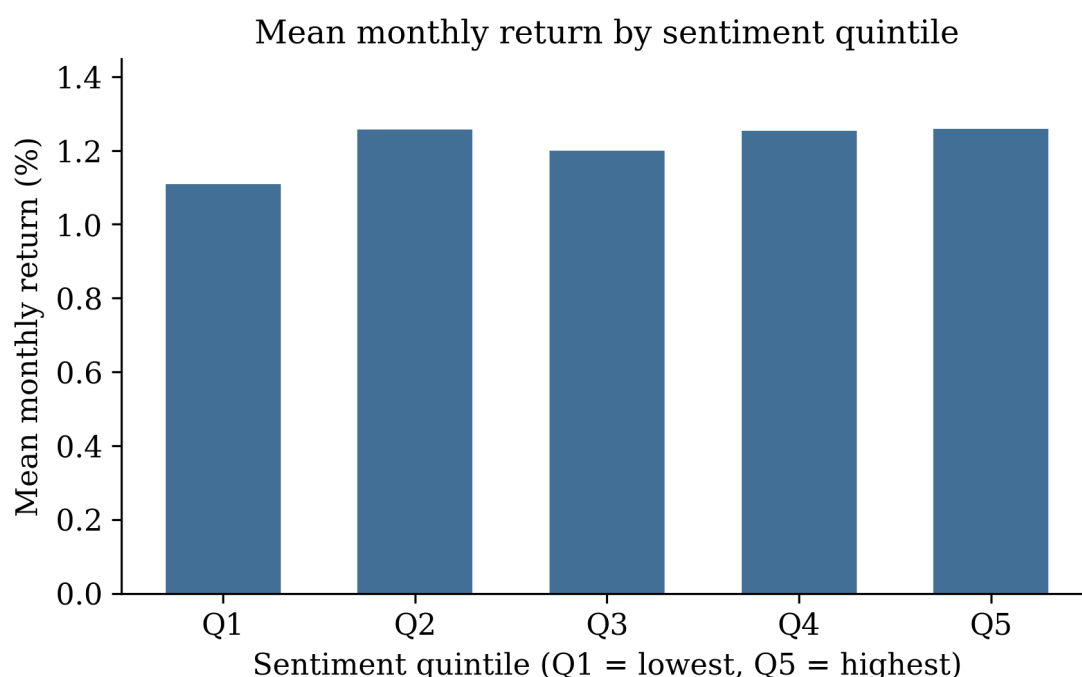


Figure 1. *Average monthly return by employee-sentiment quintile (Q1 = lowest, Q5 = highest), 2013–2023. The means are 1.11, 1.26, 1.20, 1.25, and 1.26 percent per month for Q1 through Q5; the Q5 minus Q1 spread is 0.15 percent ($t = 1.04$).*

The spread is small and highly uncertain. The long–short portfolio yields 0.15 percent per month from 2013 to 2023, with a t-statistic of 1.04. A t-statistic measures the estimated return against its own noise, and as a general rule t-statistic values below about 2 are regarded as being indistinguishable from zero. It follows that the spread observed is not strong enough to allow the conclusion that it represents a reliable return premium. In terms of raw returns, high-sentiment firms did not consistently beat low-sentiment firms over the period under examination. When the sample is extended to include 2024, the same conclusion is reached: over the full 144-month period the average spread is again approximately 0.15 percent per month.

Figure 2 shows how the spread changes over time, and the pattern is mixed rather than persistent. The spread drifts slightly negative in 2014 and 2015, then reverses from 2016 through 2020, at roughly a quarter to just over half a percent a month, then dips again in 2021 and more clearly in 2022, at about -0.7 percent a month, and finally reaches its best year in 2023 at nearly one percent. The 2022 dip is consistent with that year’s rate-driven selloff, which fell hardest on the growth-leaning style the spread loads on, a style swing the risk-

adjusted tests below account for. Figure B1 in Appendix B shows the same quintiles year by year and shows that no ordering of the five groups holds together for long in either half of the window. The next section therefore examines if anything survives risk adjustment.

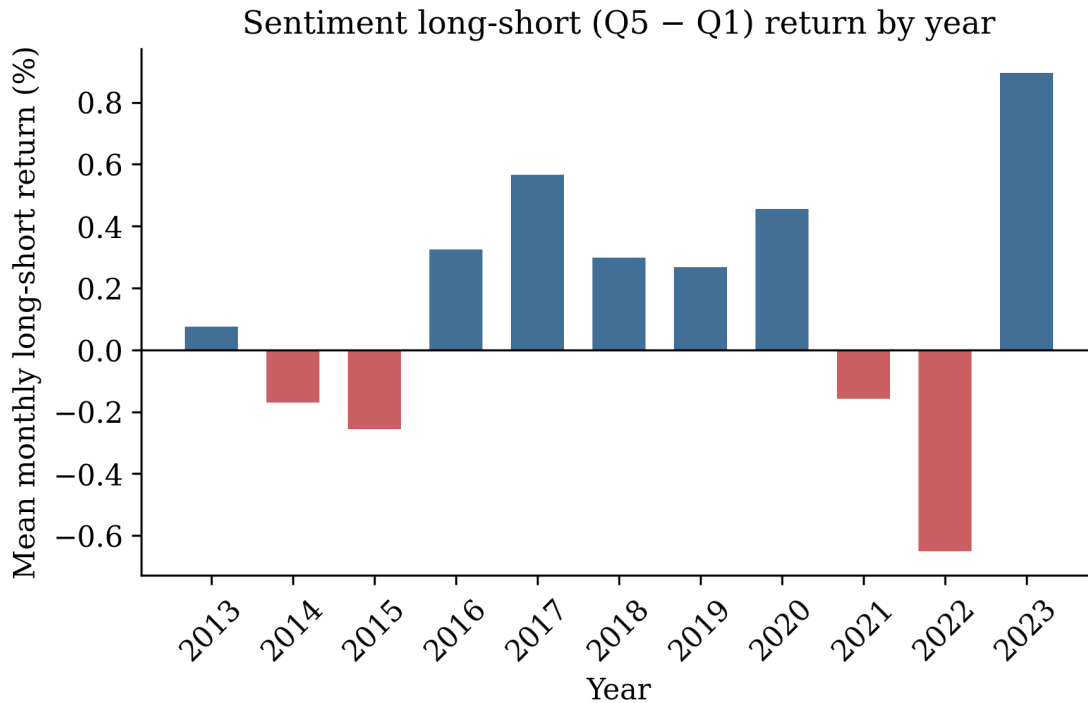


Figure 2. Long-short (Q5–Q1) spread return by year; positive (blue) and negative years (red).

4.3 Is the Premium There After Risk Adjustment? (H2a)

A difference in raw returns does not necessarily represent a sentiment premium. High sentiment firms may simply be the kind of stock that earns more or less for reasons unrelated to sentiment, such as being large, cheap, or profitable. The alpha regression removes the part of the spread return that those well-known factors explain, leaving the alpha (the piece attributable to sentiment itself). This thesis uses the five Fama and French factors plus momentum, six in all, and reports the result in Table 3.

Table 3. Risk-adjusted alpha of the long-short sentiment spread (six-factor model, Newey–West standard errors). Annualized alpha in percent; SMB is the size factor loading.

Window	Alpha (%/yr)	t	SMB loading	t (SMB)
Full, 2013–2023	1.43	1.37	–0.23	–3.59
2013–2019	0.46	0.34	–0.24	–2.99

Window	Alpha (%/yr)	t	SMB loading	t (SMB)
2020–2023	3.32	2.12	−0.20	−2.09

Note to Table 3. The full-window regression’s remaining loadings are MKT 0.05 ($t = 1.57$), HML -0.09 ($t = -1.88$), RMW -0.16 ($t = -2.11$), CMA -0.20 ($t = -2.59$), and MOM -0.08 ($t = -1.74$), estimated on the same 132 months; R^2 is 0.35, and annualized alphas are the monthly intercept times twelve. The sub-period rows report the alpha from the same six-factor regression estimated within each window. Their remaining loadings are MKT 0.04 ($t = 1.02$), HML 0.00 ($t = 0.06$), RMW -0.02 ($t = -0.23$), CMA -0.29 ($t = -2.68$), and MOM -0.04 ($t = -0.49$) for 2013–2019, and MKT 0.07 ($t = 1.45$), HML -0.13 ($t = -2.20$), RMW -0.23 ($t = -2.08$), CMA -0.14 ($t = -1.60$), and MOM -0.09 ($t = -2.33$) for 2020–2023. Under six- and twelve-lag Newey–West windows the full-window alpha’s t-statistic is 1.43 and 1.32, so the null does not depend on the lag choice.

Over the full window, the alpha is 1.43 percent a year with a t-statistic of 1.37 ($p = 0.17$), making it statistically indistinguishable from zero. The corresponding confidence interval ranges from -0.61 to $+3.46$ percent a year. This means that the evidence is compatible with anything from no premium at all to one close to the estimate reported by Edmans. Edmans’s 3.5 percent sits at the very top of that interval, technically outside it but not by a margin worth leaning on, and the one-to-two percent residue the decay literature would predict remains well within the range of plausible values. In terms of statistical power, with a standard error near one percent a year, a true alpha of 1.5 percent would be detected in only about three times out of ten, and an alpha of about 2 percent would have roughly an even chance of reaching conventional significance. The design is therefore capable of detecting a relatively large premium but is limited in its ability to distinguish a modest remaining effect from zero. This null should therefore be interpreted as evidence against a large premium rather than no premium at all, and the discussion chapter handles what it can and cannot support.

The factor loadings are more informative than the alpha itself. The spread’s strongest exposure is to size. Its SMB coefficient is -0.23 with a t-statistic of -3.59 , which means that the high-sentiment side of the portfolio behaves like large firms and the low-sentiment side like smaller ones. This reflects the modest positive size correlation from Table 2, which becomes more pronounced as the sort isolates the top and bottom quintiles. The portfolio has

significant negative loadings on the investment and profitability factors ($t = -2.59$ and -2.11), with value marginally negative as well (-0.09 , $t = -1.88$), so in factor terms the strategy leans toward large growth firms and away from the cheap, profitable, low-investment ones. Table B1 in Appendix B displays the underlying characteristics: average market equity increases from about \$28 billion in the lowest quintile to \$97 billion in the highest, book-to-market declines from 0.42 to 0.32, and return on assets rises from 5.8 to 7.7 percent. That last figure goes against the profitability loading rather than with it, serving as a reminder that a loading shows how the spread moves together with a factor return, not the average characteristic of the firms included in it. Together, the six factors account for just over one-third of the month-to-month variation in the spread. The loadings describe what the sort captures; they do not explain why there is no statistically significant premium. The raw return spread was already statistically flat, and the factor compensation accounts for only about one-fifth of it, so the alpha remains indistinguishable from zero.

The sub-periods do not change the verdict. Splitting the sample at the start of 2020 returns an alpha of 0.46 percent a year before 2020 ($t = 0.34$), as close to zero as estimates come, and 3.32 percent a year after ($t = 2.12$). Although the post-2020 number is nominally significant, it should be interpreted cautiously: the split was imposed after the data were observed, the window contains only 48 monthly observations, and the estimate would not remain significant under a stricter threshold of the kind Section 4.5 applies to the panel. The t -statistic also depends on the standard-error correction: under ordinary or White standard errors the same estimate gives $t = 1.16$ and $t = 1.34$. The formal stability tests point to the same conclusion, with one twist. A Chow (1960) test, at the 2020 split, does not reject a stable alpha (alpha shift $t = 1.25$, $p = 0.21$; joint break $F = 1.42$, $p = 0.20$). However, a search for an unknown break date rejects the stability at the five-percent level (sup-Wald 10.61 against the five-percent critical value of 8.85; Andrews, 1993), identifying October 2015 as the most likely break point. The estimated alpha is higher after this date. Since the statistic clears the five-percent critical value but not the one-percent one, is based on only 132 monthly observations, and clears that critical value only under the autocorrelation-corrected standard errors (4.06 under ordinary and 6.39 under White standard errors, both well below 8.85), the evidence should be read as suggestive rather than conclusive. Most importantly, the estimated break runs in the opposite direction from what the anomaly-decay hypothesis predicts. Figure 3 compares the three alphas side by side, while Figure 4 shows the cumulative performance of one dollar in the long-short portfolio: down to roughly 95 cents

by late 2015, a climb to about \$1.20 by 2021, a decline during 2022, and a recovery to about \$1.20 by the end of 2023.

The decline during 2021–2022 visible in Figure 4 can largely be explained by factor exposures. During those 24 months the portfolio obtained an average return of minus 0.40 percent per month, which is about 0.52 points less than the level suggested by its long-run alpha. According to the estimated loadings, the factor movements during the period in fact imply an even greater expected decline, explaining approximately 140 percent of the shortfall. The main contribution to this effect comes from the investment, profitability, and value factors (with monthly contributions of about -0.30 , -0.21 , and -0.19 percent respectively), while size adds nothing significant ($+0.02$). Although the residual alpha over the sub-period is positive it is imprecise, amounting to about 3 percent per year with a t-statistic of 0.84 and thus is not significant at all. The temporary decline is therefore due to the sharp value-and-quality rotation that took place between 2021 and 2022 rather than to a change in the underlying sentiment effect. The interpretation of this point is further discussed in Section 5.3.

A further check, carried out following the external review and thus outside pre-committed plan, involves sorting the portfolio using value weights rather than equal ones. The spread of returns for the value-weighted approach is greater. The raw long–short portfolio yields an average of 0.45 percent per month ($t = 1.90$) and achieves an annual six-factor alpha of 3.5 percent, with a t-statistic of 1.76. The t-statistic is increased only slightly when six- and twelve-lag Newey–West standard errors are used, the values going up to 1.82 and 1.91 respectively, so that the estimate still remains marginal and not significant. The value-weighted strategy also displays a more evident large-cap tilt (an SMB loading of -0.32), and this tilt is concentrated rather than widespread: the ten largest holdings account for an average of 56 percent of the weight in the long position over the same period, as compared with 40 percent in the short position and 11 percent under equal weighting. Boustanifar and Kang (2022) discount their own value-weighted estimate for the same reasons.

Since this specification was introduced after the initial research design and still falls short of conventional statistical significance, it cannot overturn the main result. Nevertheless, it indicates that the equal-weighted null does not automatically extend to every portfolio construction, an issue revisited in Section 5.5 when comparing the present findings with the value-weighted evidence in Edmans (2011).

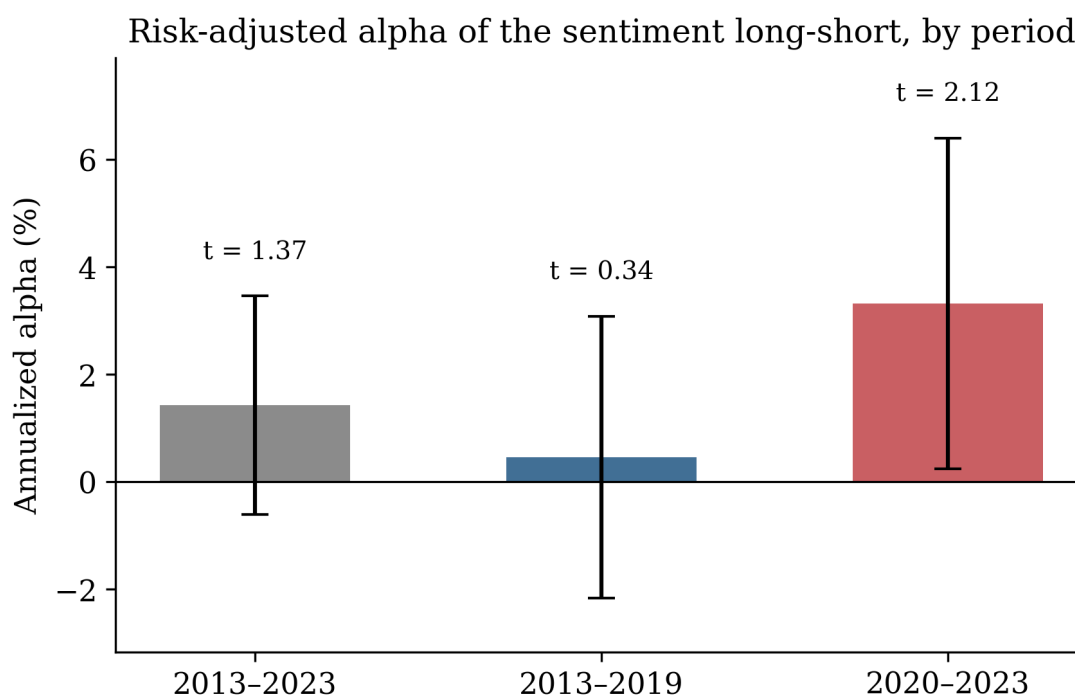


Figure 3. Annualized alpha for the full window and the two sub-periods, with 95 percent confidence intervals.

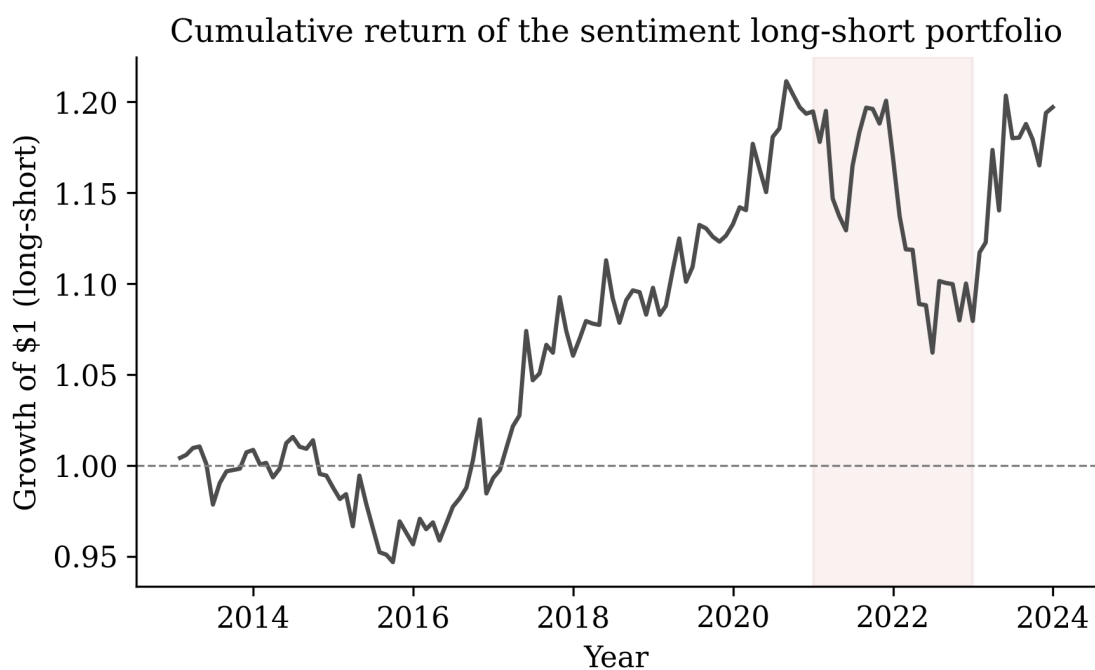


Figure 4. Cumulative value of one dollar in the long-short sentiment portfolio, 2013–2023; 2021–2022 shaded.

4.4 Does Sentiment Predict Operating Performance? (H2b)

Whatever the market does with the information, sentiment might still predict the firm's own results. Table 4 reports panel regressions of each year's operating outcome on the prior year's sentiment, controlling for firm size and leverage, with firm and year fixed effects. The firm fixed effects do the work here: every company is compared only against its own history, so the estimate cannot come from stable differences between industries, and the year fixed effects absorb anything that moved all firms at once.

Table 4. Panel regressions of year- t operating outcomes on year- $(t-1)$ sentiment, with firm and year fixed effects and standard errors clustered by firm; outcomes winsorized at the 1st and 99th percentiles, with firm-years missing an outcome dropped before winsorizing. Within-firm R^2 is 0.016 for sales growth, 0.033 for return on assets, and 0.040 for operating margin.

Outcome	Coefficient	t	p	N
Sales growth	0.0211	1.80	0.073	4,954
Return on assets	0.0038	1.68	0.093	4,956
Operating margin	0.0009	0.22	0.83	4,725

No outcome clears the conventional line. The point estimates lean the expected way, but the evidence is thin. A one-point rise in a firm's sentiment score is associated with a 2.1-percentage-point increase in next-year sales growth, yet the estimate is only marginal ($t = 1.80$, $p = 0.07$). Return on assets moves with sentiment by about four-tenths of a percentage point and falls short as well ($t = 1.68$, $p = 0.09$). Operating margin shows no relationship at all ($t = 0.22$). Its smaller sample reflects the financial firms whose records lack the operating-income item. The estimated coefficients are also economically modest. Since a full one-point move on a five-point scale is quite large, a more realistic one-standard-deviation improvement in sentiment, about half a point, represents roughly one percentage point of sales growth. Overall, the panel provides only suggestive evidence for the sales-growth channel and no significant effect anywhere.

The pattern across the three outcomes is nevertheless informative. The strongest relationship appears for sales growth, a measure of how much business a firm does; return on assets, which captures profitability relative to assets, responds less; and operating margin, the marker of pricing power, not at all. Taken together, these results are more consistent with employee satisfaction supporting higher sales volumes than greater pricing power, the same channel the

SHRM literature describes. The robustness checks next determine whether this pattern persists under alternative specifications.

4.5 Robustness

Table 5 reports the robustness checks, with operating margin omitted because the headline estimate is null. The first three rows follow pre-committed specifications, the falsification row a pre-specified one, and the two sub-period rows are descriptive context rather than planned tests (Section 3.5). Overall, the robustness checks refine the interpretation of the findings rather than strengthening them. As several of them run the same kind of test many times over, the risk of false positives increases. Harvey et al. (2016) therefore recommend a stricter threshold at a t-statistic of about 3.0, a standard applied uniformly below.

Table 5. *Robustness checks for the operating-performance results, 2013–2023; cells show the sentiment coefficient with its t-statistic and sample size. Rows one–three are pre-committed; the falsification row is pre-specified; and the sub-period split is descriptive.*

Specification	Sales growth (t)	ROA (t)
Headline	0.0211 (t = 1.80, N = 4,954)	0.0038 (t = 1.68, N = 4,956)
At least 30 reviews	0.0176 (t = 1.19, N = 4,042)	0.0011 (t = 0.23, N = 4,043)
Precision-weighted ($\sqrt{n}$)	0.0264 (t = 1.78, N = 4,954)	0.0022 (t = 0.48, N = 4,956)
Falsification (backward)	−0.0106 (t = −0.98, N = 3,724; passes)	0.0075 (t = 2.13, N = 3,725; fails)
2013–2019 sub-period	0.0291 (t = 2.29, N = 3,070)	0.0051 (t = 1.88, N = 3,071)
2020–2023 sub-period	0.0152 (t = 0.47, N = 1,884)	−0.0001 (t = −0.02, N = 1,885)

The measurement checks distinguish between the two outcomes. Restricting the sample to firm-years with at least 30 underlying reviews drops nearly a fifth of all observations, mainly the thin years from the early part of the study. For sales growth, the coefficient declines by roughly a sixth, but the estimate is no longer marginally significant (t = 1.19). This change likely reflects a mix of lost power and a thinner signal that the data cannot separate. By contrast, return on assets does worse: its coefficient drops by about two-thirds (t = 0.23) and becomes effectively identical to zero. In the second measurement check the full sample is kept but each firm-year is weighted by the square root of its review count, so that firms with

better data count more. Under this specification, sales growth maintains its marginal significance ($t = 1.78$) while return on assets again shows no evidence ($t = 0.48$). Together, sales growth acts as a weak signal that is sensitive to measurement noise, while return on assets behaves as if there is no signal at all.

The falsification test leads to the same conclusion. It examines whether sentiment appears to predict the past as well as the future, as mentioned in Section 3.5. In case last year's sentiment is just as related to performance two years ago as to performance next year, the forward result may reflect persistent co-movement rather than predictive power. The falsification samples are smaller than the headline's because the outcome measured two years back has to lie inside a firm's membership window, which excludes the panel's first two years and any gap years. Sales growth does pass cleanly, with no backward relationship (backward $t = -0.98$). Return on assets fails: prior-year sentiment is related to return on assets two years before the outcome year at least as strongly as to next-year return on assets (backward $t = 2.13$). Return on assets is highly persistent within firms, and sentiment and profitability appear to move together over multi-year periods. That relationship is therefore best interpreted as long-run co-movement rather than prediction.

The interpretation is also dependent on the purpose of the statistical threshold here. Harvey et al.'s stricter benchmark of about 3.0 is intended for new empirical findings, and nothing in this thesis clears it: neither the sales-growth association ($t = 1.80$), nor the sub-rating estimates below (t up to 2.39), nor the post-2020 alpha ($t = 2.12$). Nevertheless, the falsification check serves a different purpose. It is designed to detect evidence against a predictive interpretation rather than to establish new findings. Consequently, the forward reading needs the backward association to be absent, so a backward t of 2.13 is disqualifying even though it would not count as a finding. Each of these asymmetries works against the results of this thesis: the one significant return estimate is discounted, the sub-ratings are not treated as discoveries, and return on assets is downgraded because of a marginal backward association.

One final convention should be noted. All the p -values reported are two-tailed, even though the predictions in Section 2.4 are directional. If a one-tailed test is used, sales growth would clear the five-percent level ($p = 0.036$). Yet the more stringent two-tailed convention is maintained throughout.

The sub-rating breakdown no longer identifies any clear favorites. When the panel is re-run with each of the five sub-ratings in place of the overall score, no dimension predicts return on assets at the five-percent level (senior leadership comes closest at $t = 1.74$), while four of the five predict sales growth at similar strength: compensation ($t = 2.39$), culture ($t = 2.37$), senior leadership ($t = 2.36$), and work-life balance ($t = 2.15$), with career opportunities the only null ($t = 0.97$). Figure 5 displays the estimates. The sub-ratings are highly correlated with one another, since a firm rated well on leadership tends to be rated well on culture too, so no ranking should be read into these numbers. What they show is that the sales-growth association is broad-based across the dimensions rather than concentrated in any one of them. The sub-rating samples are nearly the full panel (N between 4,938 and 4,947, against 4,954), and the overall score is an item reviewers rate directly rather than an average of the five, so the slightly stronger sub-rating statistics are not a sample artifact. The overall rating and its individual dimensions should therefore be viewed as related but distinct measures of workplace quality. The findings of this thesis are different from those of Green et al. (2019). Those researchers found a more distinct hierarchy, with senior management showing the strongest association and work-life balance showing none, but their approach is also too different for the contrast to be of any significance here. They relate changes in ratings among current employees on a quarterly basis to stock returns, while this study uses the annual rating levels from all the core reviewers and compares them with the operating results. Given these substantial differences in data and measurement, the contrast should not be interpreted as evidence for or against either set of findings.

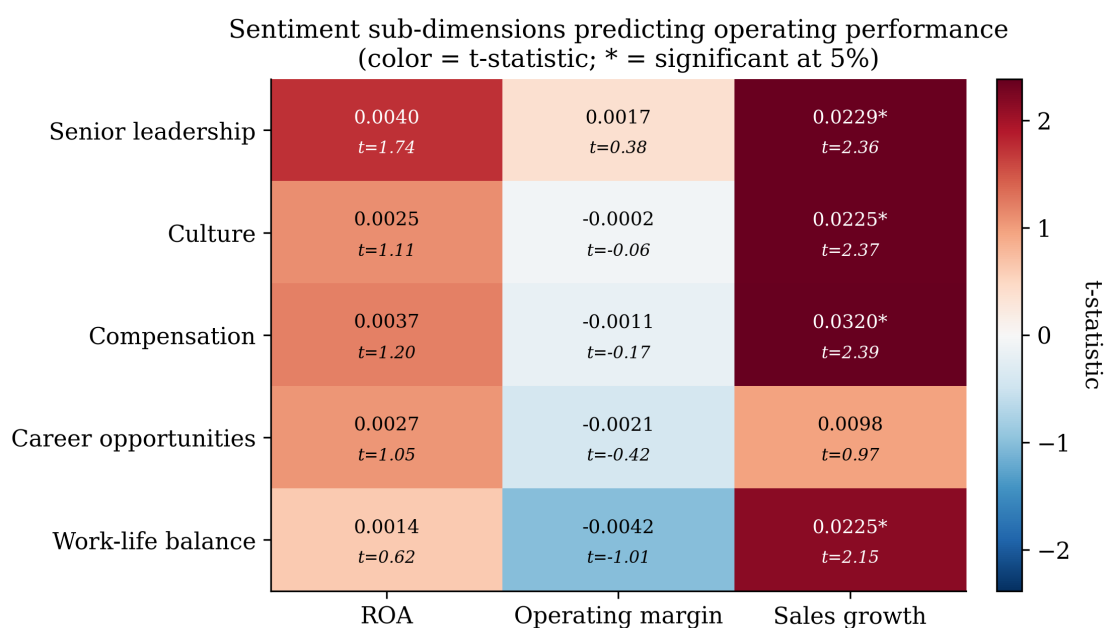


Figure 5. *Predictive strength of the five sub-ratings across the operating outcomes, colored by t-statistic; asterisks denote significance at the five-percent level, before the stricter threshold applied throughout this section.*

Splitting the panel at 2020 adds a final qualification to the results. The association between sales growth and sentiment is concentrated in the 2013–2019 period ($t = 2.29$, read descriptively and with the same reservations the post-2020 return alpha receives). The connection is absent from 2020 on ($t = 0.47$), just like the return on assets, which is marginal before ($t = 1.88$) and absent after, as Table 5 reports. The operating performance link lives in the first seven years. Interestingly, the asset-pricing estimates in Section 4.3 show a different pattern. While the operating-performance link weakens after 2020, the return alpha is somewhat stronger during the after period. The two relationships therefore do not evolve in parallel, indicating that the mechanisms that link employee sentiment to operating performance and stock-market performance may differ. Chapter 5 discusses this asymmetry in more detail.

Taken together, the robustness checks lead to a rather cautious interpretation. Employee sentiment does not earn reliable risk-adjusted return over the 2013–2023 sample, nor within any sub-period once the limitations of the post-hoc split, the 48-month window, and the stricter bar of Harvey et al. (2016) are considered. The operating-performance results provide somewhat more encouraging evidence, but remain limited. The positive findings rest on a marginal sales-growth association that is concentrated before 2020, spread across most sentiment dimensions, and sensitive to how well sentiment is measured. In contrast, the return on assets results fail both the measurement and falsification checks, making a predictive interpretation difficult to support.

5. Discussion

5.1 Synthesis

The central question of this thesis was whether employee sentiment can be used to predict firm performance, both in the managerial and financial areas. There were three possible results: it might have been able to predict both operating results and returns, it might have predicted operating results only, or it might have predicted neither. The evidence presented in this thesis is most compatible with the second of these outcomes, although not fully. The sales-growth relationship, although weak, comes from a pre-committed test, passes the falsification test, and remains positive after the robustness checks have been carried out. The

relationship to return on assets fails that test, so the operating results rest on sales growth alone. On the other hand, the only return estimate that attains conventional significance is based on a post-hoc split and uses a 48-month estimation period. Even so, neither result should be read as a discovery. By the standards Section 4.5 sets out, none of the results clear the stricter threshold applied throughout this thesis. Higher sentiment is associated with faster sales growth the following year, but this association is marginal, becomes weaker when the noisiest observations are excluded, and is mainly observed before 2020. The return tests, meanwhile, find no premium that withstands the checks applied in Chapter 4. Worker wellbeing might therefore be a real business asset, even if its impact on firm performance seems to be modest, and there is no significant evidence that it led to abnormal stock returns between 2013 and 2023. The results are consistent with the premium having been priced in before the sample window begins, although the design cannot pinpoint when exactly that happened, and the sub-period evidence points in the opposite direction.

The findings belong to the firm-level literature following Edmans (2011), which asks whether markets fully value a workforce characteristic. A different question is whether changes in economy-wide employee sentiment help predict overall market returns. Chen et al. (2023) take that question up and find a negative relationship: high economy-wide employee sentiment forecasts lower future market returns. This thesis, by contrast, looks into whether companies with more satisfied employees outperform those with less satisfied employees after accounting for risk. The evidence from 2013 to 2023 shows that they do not. Neither the analysis of the sub-period nor the weighting variant offers enough evidence to contradict that conclusion, considering the limitations mentioned in Chapter 4. Two principles apply to the discussion that follows. The first concerns status: the 2020 split was not part of the pre-committed design but was chosen after the return series had been observed and was a response to major economic events, so its results are treated as descriptive background information. The second concerns inference: wherever the data appear to show the effect changing across the window, the short sub-samples and the range of break dates that could have been examined make noise the default reading. The discussion starts with the operating results before moving on to the returns.

5.2 Employee Sentiment and the Firm's Own Performance

The operating evidence points in the right direction yet is still not conclusive. If employee sentiment increases by one point on the five-point scale, sales growth the following year is

about 2.1 percentage points higher. A more realistic increase of one standard deviation in sentiment, roughly half a point, corresponds to about one percentage point of growth. The estimate is only marginally significant ($t = 1.80$, $p = 0.07$) and is further examined in the robustness checks of Section 4.5. Its statistical significance is almost the same when precision weighting is applied ($t = 1.78$ instead of 1.80; although the coefficient increases by about a quarter), but it loses even its marginal significance when the firm-years that have been thinly reviewed are excluded. The association is also mainly found in the years 2013 to 2019. The firm fixed effects (Section 3.4) mean the estimate reflects changes in sentiment within firms over time, not differences between industries.

Compared with the SHRM literature, the finding is basically a weaker version of a strongly established pattern. Huselid (1995) found that firms which adopt more effective people-management practices see both a reduction in staff turnover and an increase in productivity, and Combs et al. (2006) have confirmed this pattern through a meta-analysis (Section 2.1). Jiang et al. (2012) provided an explanation of the mechanism: HR systems affect financial performance both directly and indirectly via the workforce, namely by making employees more skilled and more motivated. In this connection, a sentiment score represents the observable outcome of a wider process, and the positive direction of each estimate in this thesis is in agreement with that literature. What this sample is missing, though, is the expected strength the literature would predict. Part of the difference could be due to methodological factors: since crowd-sourced sentiment is a noisy stand-in for the practices that those studies measured directly, such mismeasurement naturally leads to a weakening of the estimated relationships. A different part of the explanation may also be economic: among the largest, most professionally managed firms in the world, the differences in employee treatment may simply be too small to produce large differences in firm-wide performance.

The pattern across the estimates is still informative. Sales growth shows the strongest association with sentiment, return on assets a weaker one, and operating margin none at all. This pattern is more consistent with higher business activity, the mechanism Huselid (1995) documented, than with greater pricing power. The sub-rating breakdown, however, does not identify a single driver. Not a single dimension stands out: four of the five sub-ratings predict sales to a similar degree, work-life balance being among them, while none predict return on assets. The resource-based view (Barney, 1991) would instead suggest that the hard-to-imitate dimensions leadership and culture should matter most while easily matched amenities

should matter less. Green et al. (2019) found exactly that pattern for stock returns, with changes in senior management having the strongest predictive power. This thesis's sample does not reproduce that ordering. The two designs differ enough, though: levels rather than changes, all core reviewers rather than only current employees, operating outcomes rather than returns. Given the strong correlations among the dimensions, the only defensible conclusion is that the data cannot isolate which aspect of the employee experience is responsible for the association.

The case for profitability is even weaker. Even though there is a positive relationship between sentiment and return on assets as shown in the records, this relationship fails the falsification test. Last year's sentiment is linked to return on assets two years before the outcome year at least as strongly as to next year's, which is not how a forward-looking signal should behave. Profitability is strongly persistent within firms, and sentiment appears to move alongside it over longer periods of time rather than predict future changes. The operating-performance result therefore rests on the sales growth result alone, marginally, while the return on assets relationship is better interpreted as long-run co-movement rather than as prediction.

5.3 The Asset-Pricing Result

The return results are easier to summarize. In the main specification, there is no reliable premium to explain. The long-short portfolio earns a risk-adjusted alpha of 1.43 percent a year, with a t-statistic of 1.37, well within the range of sampling noise. Nevertheless, the spread does have a clear style pattern: its loading on the size factor is -0.23 , a figure that is estimated accurately ($t = -3.59$), and its loadings on the investment and profitability factors are also negative. Taken together, this indicates that the high-sentiment side of the trade acts like large, capital-intensive growth companies while the low-sentiment side is similar to smaller, cheaper firms. The sentiment-size correlation of just 0.10 shown in Table 2 becomes a real exposure at the two extremes of the trade. The factor loadings are not responsible for the null result; in fact, the raw spread was statistically flat before any adjustments were applied, and factor compensation accounts for about one-fifth of the return. It has for a long time been difficult to tell an actual sentiment effect apart from well-known firm characteristics, even when giving the effect a different name; Daniel and Titman (1997) claim that the average returns might be due to firm characteristics rather than to the factor loadings, and therefore a factor model on its own would not be able to fully separate the two. In this case, the distinction matters little, since there is no reliable return to attribute to either

reading. A stronger test would compare firms with similar characteristics, of the kind Edmans (2011) also reports, which is not run here. The value-weighted variant of the spread remains a separate case, marginal and post-hoc (Section 4.3), and Section 5.5 weighs what it can and cannot say.

The more interesting part is what did not occur. The modern-era literature reviewed in Chapter 2 suggested that any satisfaction premium should weaken over the given period. McLean and Pontiff (2016) show that published return anomalies lose more than half their strength once investors learn about them, while Pástor et al. (2022) argue that rising investor demand for a stakeholder-linked characteristic raises prices today and trims expected returns in the future. Employee sentiment between 2013 and 2023 fits both arguments almost perfectly, since Edmans's evidence was published in 2011 and crowdsourced ratings became abundant roughly at the same time. The natural prediction was a premium that was stronger during the early part of the window and faded over time.

The data do not indicate this at all: the alpha during the period 2013–2019 is 0.46 percent per year ($t = 0.34$), which is effectively zero, and the estimate for the period 2020–2023 is higher, not lower, at 3.32 percent ($t = 2.12$). A Chow test, at the pre-determined break point of 2020, fails to reject the hypothesis of a stable alpha. A search of the unknown break dates does find evidence of instability at the five-percent level, but the estimated break occurs in October 2015 and is associated with a higher, rather than lower alpha. Both results should be interpreted cautiously. The post-2020 estimate is based on only 48 monthly observations, and the break date search was conducted after the fact using significance thresholds that become less convincing once multiple testing is considered (Harvey et al., 2016).

The most reasonable conclusion to draw is therefore modest and informative: there is nothing in this sample which looks like a premium gradually vanishing as investors become aware of it. The magnitude of the full-sample estimate is also consistent with the decay literature: McLean and Pontiff report an average post-publication decline of 58 percent, and applied to Edmans's 3.5 percent, that leaves roughly 1.5 percent a year. The estimate over the full window in the present case is 1.43 percent, which is no more different from that residual amount than it is from zero. The comparison should not be taken literally, though, since the two designs differ enough in their settings, but it still provides a useful benchmark. At the same time, Section 4.3's power arithmetic shows that the present design does not reliably distinguish such a residue premium from nothing.

The interpretation most consistent with both the theory and the evidence is that any market learning had already taken place before the sample provides enough information to detect it. If the market slowly came to price employee satisfaction following the time when Edmans (2011) made the signal well known and Glassdoor made it visible, then a large part of that adjustment would almost certainly have taken place in the early and middle 2010s. As discussed in Section 2.2, Edmans (2012) stated that it could take markets four to five years to fully incorporate intangible information, which means that any learning process should have been essentially completed by about 2015–2016. The one break the stability search identifies does fall in October 2015, close to where that arithmetic would put the end of learning. The coincidence is worth recording rather than relying on, since the search is post-hoc and the shift runs in the wrong direction. It is also rather strange that the alpha increases afterwards, exactly the opposite of what one would expect if learning had by then been completed. Although the timing is consistent with the hypothesis, the direction is contrary to it. This thesis therefore points out this contradiction rather than dealing with it. The design is unable to reliably tell the difference between a premium that had disappeared before the sample period, one that disappeared during the early years of the sample, or one that never existed among large and publicly traded firms with broad analyst coverage and genuinely limited investor neglect.

A few loose ends still have to be attended to. First, the positive alpha observed during the pandemic period is handled with care, since 48 months of an equally weighted spread is precisely the situation in which style noise pretends to be alpha. Second, the dip between 2021 and 2022 is minor and can be explained: as Section 4.3 states, the value-and-quality rotation during that period explains the entire dip through the negative value, profitability, and investment loadings of the spread, size plays essentially no role, and although the 24-month residual is positive it is also noisy, being the same weak pandemic-era residue that the previous paragraph chooses not to interpret. The reason for the dip was style exposure and not a failure of the signal.

The operating results and the returns also sit in different parts of the window. The link between sales growth and sentiment during the period in question is concentrated before 2020, while the return estimates are most prominent afterwards. For whatever reason connects workforce sentiment to firm value, the operating and pricing channels do not move up and down at the same time over this period. No single account based on anomaly decay,

market incorporation, attenuation, or structural change fits that pattern neatly, though with both channels this weak the difference in timing may itself be noise.

5.4 Limitations and Threats to Inference

These results should not be overstated in either direction. The main finding is based on a marginal estimate since its headline t-statistic of 1.80 is below the conventional level for significance. If the sample is limited to firm-years that have 30 or more reviews, nearly one fifth of the observations are eliminated and both the coefficient is reduced by about one sixth and the t-statistic is lowered, so that it becomes impossible to tell whether the low statistical power or a weaker true signal is responsible. Return on assets does even worse, failing both the measurement and the falsification tests. The sub-ratings are also too intercorrelated to identify which aspect of sentiment matters most, and the borderline results change across specifications. Harvey et al. (2016) argue that once many predictors are tested, the threshold should rise to a t of about 3.0; by that standard, nothing in this thesis, on either channel, would count as a discovery. The fair summary therefore is that the estimates point in a direction, but none of them establish it convincingly.

The sentiment measure has its own limits. As already mentioned in Section 3.6, crowdsourced reviews are written by employees who choose deliberately to do so, and as a result they do not always give a true reflection of the whole workforce, and both the people who post them and the way the vendor collects and scores reviews have shifted over the decade. The unevenness in the coverage is considerable, since a small number of companies could not be associated with any review data and several dozen were never reviewed at all. The number of reviews increased sharply, rising from a median of a few dozen in the early years to over three hundred by 2021, so the data from the early years are measured much more noisily than that from later years. This applies in both directions. Random noise in a predictor tends to drive the estimates towards zero, so the marginal operating estimate is more likely to be understated than overstated. The return null is the more difficult case. The same kind of noise could have reduced a real premium down to the point where this design can no longer detect it, and the power calculations in Section 4.3 acknowledge this. It also means, though, that the pre-2020 concentration of the operating result rides on the noisiest stretch of data.

Sample construction is another place where caution matters. The panels admit a firm's outcome years only while the firm actually belonged to the index. The comparison against the

pre-correction panel described in Section 3.6, which included ever-members' full histories, a common shortcut in index studies, shows what that shortcut costs: those histories include the pre-inclusion years of later entrants, systematically strong performers, and with them included the panel shows a marginal pre-2020 return premium, an annualized 2013–2019 alpha of about 3.0 percent ($t = 1.83$) that point-in-time membership cuts to 0.46 percent ($t = 0.34$), with the full-window alpha barely moving (1.33 against 1.43 percent a year). Look-ahead of that kind is easy to miss, roughly a fifth of the panel here, and it spans the difference between an apparent premium and none.

In short, the results are specific to a particular setting: the analysis covers the largest U.S. firms over a single window from 2013 through 2023, and the null returns and the weak operating link need not extend to smaller firms, firms in other markets, or other periods. The international evidence provided by Edmans et al. (2024) indicates that the effect varies according to labor-market institutions. The operating test accounts for fixed effects and has a lag structure; it only establishes an association, not a causal relationship. Additionally, the fiscal-year alignment between calendar-year sentiment and fiscal-year outcomes is imperfect for a minority of firms whose fiscal years do not end in December, about a quarter of the sample. The one-year lag reduces, but cannot completely eliminate this mismatch. None of these limitations overturn the main finding, but they do set the terms on which it can be read.

5.5 Contributions, Implications, and Future Work

Each of the contributions should be read in the light of the limitations discussed above. The first is the test itself: the employee-satisfaction premium in the broad modern S&P 500 that is measured from continuous crowdsourced reviews rather than a curated list cannot be detected at conventional levels. The evidence argues against a premium as large as the original one, although only narrowly, and it cannot rule out the small remaining premium that the decay literature predicts. That result sits awkwardly beside the earlier evidence. Boustanifar and Kang (2022) find that the “100 Best Companies to Work For” premium continued through 2020, while Edmans et al. (2024) find it across countries with flexible labor markets. At the same time, the comparison applies in both directions. The results of Edmans et al. (2024) show that the United States should have one of the strongest premiums on the grounds of labor-market flexibility, although the U.S. alpha they calculate places it 16th out of the 21 countries they examine. Nevertheless, the comparison makes it more difficult to explain the thesis's result solely on the basis of labor-market institutions.

There are several other explanations possible. The difference could come from the measure, the sample, the time period, or the portfolio construction. Edmans (2011) uses a value-weighted long-only portfolio design, whereas this thesis uses an equal-weighted long-short one. The marginal 3.5 percent alpha from the value-weighted check in Section 4.3 sits closer to that design than the equal-weighted headline does. Nevertheless, this interpretation is weak, since, as Section 4.3 points out, ten individual companies make up 56 percent of the value-weighted long side, and thus the estimate reflects at least as much the effect of a small number of large-cap firms as it does a satisfaction premium; this is the very reason why Boustanifar and Kang discount their own value-weighted result. The comparison does not resolve cleanly, however: Boustanifar and Kang (2022) find the Best Companies premium in the equal-weighted portfolio and do not in the value-weighted one, the reverse of the pattern in this thesis. It is therefore possible that the difference from Edmans (2011) can be accounted for by the portfolio design, but it does not explain the discrepancy with Boustanifar and Kang (2022). It is still an open question whether or not the various explanations can be distinguished. Furthermore, the finding is in agreement with the wider anomaly literature (McLean & Pontiff, 2016): once a signal becomes cheap, widely available, and easy to trade on, there should be very little premium remaining.

The second point relates to the joint research design. Since the return and operating questions are examined for the same firms and over the same period, this thesis is able to conclude that any remaining value of employee sentiment can be found in firm fundamentals at best, and in that case the evidence is only slight. This is a conclusion that neither of the two sets of tests by themselves could reach. A methodological point arises from the way the sample was constructed. The look-ahead correction disclosed in Section 3.6 and quantified in Section 5.4 turned a marginal pre-2020 premium into none at all. An error of that sort is easy to overlook, which makes it a practical warning to anyone working with alternative data in index samples.

The practical implications are straightforward. For managers, the evidence is rather suggestive than conclusive. Firms whose employees rate them more highly tend to grow a little faster, a link consistent with the HR literature, although the evidence is too weak to justify strong conclusions. Investors, on the other hand, get a duller message: no detectable return edge comes from buying high-sentiment large caps and selling low-sentiment large caps. Instead, the strategy mainly captures exposure to well-known features, firm size above all. As for researchers, two methodological lessons particularly stand out: it is important to

note the date on which a signal became publicly observable, and to construct index samples using point-in-time membership, since both decisions can determine the answer.

Several extensions would sharpen what this thesis only suggests. Adding more post-2020 years to the sample would show whether the operating link revives and whether the positive late-window return estimates are noise, as assumed here, or something genuinely new. A design that is changes-based, that links shifts in employee sentiment to shifts in performance, would complement the approach used here. It was not included deliberately, as it was decided to keep the specifications fixed before the results were seen. Expanding the sample beyond the S&P 500, where sentiment coverage is thinner but mispricing is likelier, would test whether the null is merely a large-cap phenomenon, which the incorporation results would predict. Finally, re-running the panel on the most reliably classified reviews would separate a genuine weakening of the signal from a change in the composition of reviewers. Together, they would firm up what this thesis only presents as a leaning: worker wellbeing may still matter inside the businesses, even if the market does not show a detectable reward for the information.

6. Conclusion

The thesis examined whether worker wellbeing is linked to a firm's results and stock performance. Based on the evidence assembled here, the answer is different for each question. There is no detectable premium for the stock and only very limited, cautious evidence regarding the firm. Of the three alternatives described in Chapter 1, the evidence is most in line with the second one: a signal which is genuine within the firm but already reflected in the prices. However, this is only a loose fit, since the operating link is weaker than the implication of that interpretation would allow.

The conclusion regarding asset pricing is the clearer one. From 2013 to 2023, when risk was taken into account, companies in the S&P 500 with more satisfied employees did not beat those with less satisfied employees either over the whole period or in any of the sub-periods or under any of the weighting variants that satisfied the criteria set out in Chapter 4. The raw difference in returns was already no different from zero, so risk adjustment had little left to explain. Instead, what the sort reliably produces is a style profile: larger and more growth-tilted firms on the high-sentiment side, not a premium. Nothing in the evidence points to a premium gradually disappearing over the observed period. This suggests that if employee satisfaction was ever mispriced, most of that adjustment had happened before this sample

begins, or during its earliest years. While the estimates argue against an effect as large as the original one, though not decisively, they cannot exclude the smaller remainder that the decay literature predicts.

The verdict on operating performance is more supportive of employee wellbeing, although the evidence on which it is based is weaker. Companies with higher employee sentiment tended to increase their sales a little faster the next year, about one percentage point per standard deviation of sentiment, but the estimate is marginal, weakens when noisy observations are removed, is concentrated before 2020, and spreads across most dimensions of the employee experience rather than pointing to any single one of them. The link with profitability does not pass a falsification test and should therefore be seen as a long-term co-movement between firm performance and sentiment. In general, every direction of the finding matches what the SHRM literature predicts, with effects smaller than that literature would predict.

When put together, the findings are consistent with the later stage of an alternative-data signal's lifecycle. The signal is now broadly observed and easy to obtain. It still leaves minor traces in firm fundamentals, but the market shows no excess returns for it that the present design can detect. The main contributions of this thesis have been to measure that finding using a broad modern sample with both channels included in a single design, and in the process to demonstrate how easily an index sample constructed without point-in-time membership can manufacture an apparent pre-2020 premium where a clean sample shows none.

For managers, investing in a well-run workplace remains a reasonable bet on the firm fundamentals, even though the data can only offer weak support. For investors, the easy version of the trade does not show up here, regardless of whether it ever was present. As for the question in the title, the straightforward answer is that worker wellbeing may still pay off inside the firm, while the stock market, based on this evidence, does not reward it. Whether investors learned the lesson as the data became observable, or whether they never needed it since large-cap firms have been under close observation, this design cannot answer.

Appendix A. Variable Definitions

All variables are constructed in the version-controlled pipeline described in Chapter 3; commit references are submitted as a supplementary file. WRDS source tables appear in parentheses.

Employee sentiment. Firm-year mean of the overall rating (one to five) across reviews posted in that calendar year (Revelio Labs, `sentiment_individual_reviews`), keeping reviews whose employment status is regular or blank and dropping the explicitly non-core status categories (interns, contractors, freelancers, and part-time, temporary, and seasonal staff); entered with a one-year lag. The five sub-ratings (work–life balance, compensation, career opportunities, culture, senior leadership) are firm-year means of the same construction, each optional within a review. The review count `n_reviews` records how many reviews stand behind each mean.

Monthly return. CRSP total return including dividends (`crsp.msf, ret`) for primary common-equity listings linked through the CRSP–Compustat table (`ccmxpf_linktable`), link primacy P or C and link types LU, LC, and LS. Market equity is the absolute value of month-end price times shares outstanding. Index membership is taken from year-end snapshots, so a mid-year join or exit takes effect at the following snapshot.

Operating outcomes. From Compustat North America (`comp.funda`; industrial format, standard data format, domestic population, consolidated statements; fiscal years 2011–2024): return on assets is net income over total assets (`ni/at`); operating margin is operating income before depreciation over revenue (`oibdp/revt`); sales growth is revenue over prior-fiscal-year revenue minus one, matched on fiscal year.

Controls. Log of total assets (`ln of at`) and leverage (total liabilities over total assets, `lt/at`), with leverage winsorized like the outcomes.

Risk factors. The Fama–French five factors and momentum from the Kenneth R. French Data Library, merged to CRSP months on year and month.

Winsorization. Outcomes and leverage are winsorized at the 1st and 99th percentiles at the estimation stage; firm-years missing an outcome are dropped before winsorizing, and the stored panels keep raw values. Percentiles are computed once on the full panel, and sub-period regressions reuse the same caps.

Appendix B. Supplementary Figure and Table

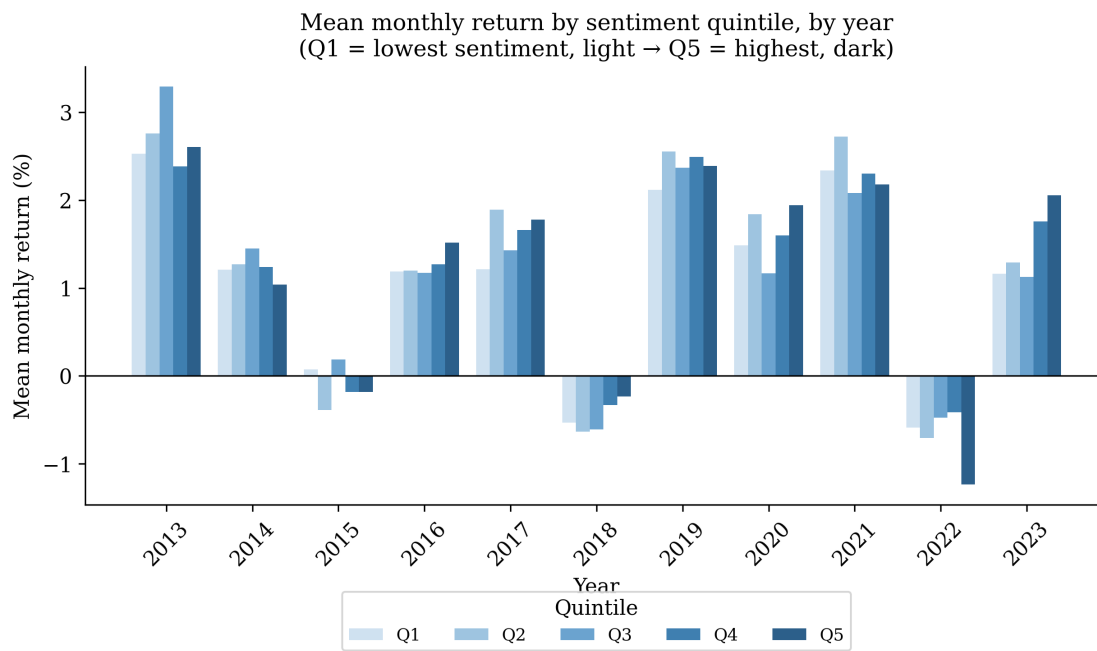


Figure B1. Mean monthly return of each sentiment quintile, year by year (diagnostic).

Table B1. Mean firm characteristics by sentiment quintile, 2013–2023.

Characteristic	Q1	Q2	Q3	Q4	Q5
Sentiment (lagged)	2.88	3.32	3.52	3.71	4.06
Number of reviews (lagged)	294	431	547	522	417
Market equity (\$ billion)	28.2	43.0	50.5	60.0	96.9
Log total assets	9.80	10.07	10.18	10.22	10.12
Book-to-market	0.42	0.39	0.38	0.33	0.32
Return on assets	0.058	0.063	0.062	0.067	0.077
Operating margin	0.246	0.242	0.242	0.251	0.276
Leverage	0.642	0.671	0.681	0.658	0.618
Sales growth	0.078	0.071	0.066	0.067	0.126

Note to Table B1. Means of raw values across the firm-years entering the sort, with quintiles assigned as in Section 3.4 (Q1 lowest sentiment, Q5 highest). Market equity is the December CRSP market capitalization in billions of dollars; book-to-market is book equity over December market equity; all other variables are defined in Appendix A. Fundamentals align

to the portfolio year. Each quintile holds 992 to 998 firm-years, of which 991 to 994 carry fundamentals.

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